



Physics-constrained graph neural networks for spatio-temporal prediction of drop impact on OLED display panels

Jiyong Kim^a, Jangseop Park^a, Nayong Kim^b, Younyeol Yu^b, Kiseok Chang^b,
Chang-Seung Woo^b, Sunwoong Yang^{a,*,1}, Namwoo Kang^{a,c},**¹

^a Cho Chun Shik Graduate School of Mobility, Korea Advanced Institute of Science and Technology, Daejeon, 34051, Republic of Korea

^b LG Display, 10, Magokjungang 10-ro, Seoul, 07796, Republic of Korea

^c Narnia Labs, 193, Munji-ro, Daejeon, 34051, Republic of Korea

ARTICLE INFO

Keywords:

Multi-layered OLED display panel
Ball drop impact test
Surrogate model
Spatio-temporal dynamics prediction
Physics-constrained MGN
Design optimization

ABSTRACT

This study aims to predict the spatio-temporal evolution of physical quantities observed in multi-layered display panels subjected to the drop impact of a ball. To model these complex interactions, graph neural networks have emerged as promising tools, effectively representing objects and their relationships as graph structures. In particular, MeshGraphNets (MGNs), state-of-the-art architectures for dynamic physics simulations using irregular mesh data, excel at capturing such dynamics. However, conventional MGNs often suffer from non-physical artifacts, such as the penetration of overlapping objects. To resolve this, we propose a physics-constrained MGN that mitigates these penetration issues while maintaining high level of accuracy in temporal predictions. Furthermore, to enhance the model's robustness, we explore noise injection strategies with varying magnitudes and different combinations of targeted components, such as the ball, the plate, or both. In addition, our analysis on model stability in spatio-temporal predictions reveals that during the inference, deriving next time-step node positions by predicting relative changes (e.g., displacement or velocity) between the current and future states yields superior accuracy compared to direct absolute position predictions. This approach consistently shows greater stability and reliability in determining subsequent node positions across various scenarios. Building on this validated model, we evaluate its generalization performance by examining its ability to extrapolate with respect to design variables. Furthermore, the physics-constrained MGN serves as a near real-time emulator for the design optimization of multi-layered OLED display panels, where thickness variables are optimized to minimize stress in the light-emitting materials. It outperforms conventional MGN in optimization tasks, demonstrating its effectiveness for practical design applications. In summary, the proposed physics-constrained MGN exhibits superior accuracy, robustness, and generalizability in drop impact scenarios compared to conventional MGN, demonstrating strong potential for real-time spatio-temporal prediction and design optimization.

1. Introduction

Multi-layered display panels, commonly used in electronic devices such as smartphones, tablets, and televisions, require rigorous testing to ensure their durability. Among the experimental procedures for evaluating the mechanical properties of these panels, the dynamic ball drop test for determining structural damages is indispensable. This test is essential for improving impact resistance by minimizing stress distribution in light-emitting materials. However, this experimental

approach is expensive and time-consuming, as it requires repeated trials and specialized instruments, such as high-speed cameras and sensors. While finite element method (FEM)-based computer simulations using explicit dynamics are widely used as an efficient alternative (Fan, Jia, Xia, Li, & Zhang, 2021; Hwan, Lin, Lo, & Chen, 2011; Liu et al., 2018; Pan, Chen, & Ho, 2003; Wu, Song, Yeh, & Wyatt, 1998; Wu et al., 2022), they are still limited in their ability to evaluate performance in real-time manner due to the high computational cost stemmed from the

* Corresponding author.

** Corresponding author at: Cho Chun Shik Graduate School of Mobility, Korea Advanced Institute of Science and Technology, Daejeon, 34051, Republic of Korea.

E-mail addresses: sbo05050@kaist.ac.kr (J. Kim), jangseop@kaist.ac.kr (J. Park), nayong.kim@lgdisplay.com (N. Kim), yyyu@lgdisplay.com (Y. Yu), kschang@lgdisplay.com (K. Chang), wcs0916@lgdisplay.com (C.-S. Woo), sunwoongy@kaist.ac.kr (S. Yang), nwkang@kaist.ac.kr (N. Kang).

¹ Co-corresponding authors.

complexity and nonlinearity of the physics. These constraints highlight the need for the innovative methods to evaluate or optimize panel designs with real-time manner.

To address these challenges, data-driven surrogate modeling has been proposed as a promising solution. It allows for the rapid prediction of quantities in complex physical systems in real-time, thereby efficiently verifying product performance (Asher, Croke, Jakeman, & Peeters, 2015; He, Kong, Ji, Xiong, & Yuan, 2024; Kang, Yang, & Yee, 2022; Lang, Wu, & Mao, 2024; Shin et al., 2023; Yang, Lee, & Yee, 2023; Yang, Vinuesa, & Kang, 2024c; Yang & Yee, 2022; Zhang & Thorburn, 2022). Building on these approaches, Kim, Kim, and Kim (2024) employed machine learning-based surrogate models, including linear regression, CatBoost (Prokhorenkova, Gusev, Vorobev, Dorogush, & Gulin, 2018), and random forest (Breiman, 2001), to predict the flexibility and durability of flexible AMOLED panels with varying thicknesses and material properties under bending and drop tests. Kim et al. (2024) demonstrated that these models could rapidly infer maximum principal strain in multi-layered display panels under ball drop impact, thereby accelerating the mechanical evaluation procedure for optimizing panel structures. Although these machine learning models excel in delivering quick and accurate predictions, they are fundamentally limited by their reliance on static, data-driven mappings of inputs to outputs (Carbone, 2022). This limitation becomes particularly critical in dynamic impact tests involving high non-linearity, complex geometries, and intricate spatio-temporal evolution. Specifically, these models struggle with scalability when applied to diverse mesh geometries, as they fail to effectively capture the spatial relationships and temporal dynamics required for accurate modeling. This lack of adaptability to dynamic loading conditions further undermines their ability to represent phenomena such as localized deformation, stress distribution, and transient motion during impact events.

Similarly, Wadagbalkar and Liu (2021) utilized artificial neural networks and decision tree regression models to predict residual velocity in projectile impact analysis. While these models incorporated dynamic characteristics by predicting residual velocity over time steps, they still relied on simplified representations of dynamics. This limitation is similar to Kim et al. (2024)'s approach. Notably, they struggle with scalability across various mesh geometries, as their framework fails to capture the spatio-temporal variations that arise when mesh configurations differ significantly in resolution, topology, or boundary conditions. Consequently, such artificial neural networks and machine learning models remain limited in their capacity to recognize and model the complex hierarchical patterns inherent in mechanical systems (Zhao, Li et al., 2024).

To overcome such pain points, researchers across various domains have been exploring data-driven surrogate models that enable spatio-temporal predictions (Tekin, Karaahmetoglu, Ilhan, Balaban, & Kozat, 2021; Wang, Cao, & Yu, 2022; Wang, Tang & Zhao, 2024; Wu et al., 2024; Zhao, Liao, Pan & Zhao, 2024). These models should be better suited for capturing both the spatial distribution and temporal evolution of physical quantities, providing more detailed and accurate predictions of dynamic responses and structural integrity in complex systems. In this context, Shao et al. (2023) developed two approaches to accelerate structural analysis from a spatio-temporal perspective: a fully connected neural network (FCNN) for predicting stress distribution across the target plate in the spatial domain, and a bidirectional LSTM (Bi-LSTM) network for predicting the stress-time sequence during bullet impact in the temporal domain. While these surrogate models demonstrated high accuracy in predicting stress, closely aligning with finite element analysis (FEA) results, the proposed model had constraints in fully capturing the dynamics. The FCNN, for example, could only predict the stress distribution at the final time step for a fixed mesh, lacking the ability to generalize across different spatial configurations. Moreover, the Bi-LSTM, despite its temporal capabilities, could only predict stress for randomly selected elements on the target plate, rather

than for the entire system. Consequently, these models faced challenges in fully capturing the dynamics and providing a comprehensive, mesh-agnostic spatio-temporal prediction.

Recently, in the mechanical system, graph neural networks (GNNs) have gained significant attention since they are inherently capable of modeling the spatial relationships in complex, irregular geometries while also tracking the temporal evolution of physical quantities (Yang, Vinuesa, & Kang, 2024b). Zhao, Li et al. (2024) highlighted that unlike convolutional neural networks (CNNs), which are inherently Eulerian and constrained to interactions within local patches or regular grids, GNNs leverage the structure of their input graph to enable flexible and arbitrary patterns of spatial interactions over any range through learned message-passing. This eliminates the need for additional preprocessing required by CNNs to handle irregular grids, making GNNs particularly well-suited for mechanical systems. Furthermore, they have demonstrated the superior accuracy of GNNs in capturing non-linear, spatial, and temporal dynamics compared to other methods, such as multilayer perceptrons (MLPs) and recurrent neural networks (RNNs), as their representation and computation align closely with learned finite element solvers.

One such approach is MeshGraphNets (MGNs), a GNN-based deep learning model proposed by Pfaff, Fortunato, Sanchez-Gonzalez, and Battaglia (2020), which accurately predict the future states of irregular meshes, thereby enabling spatio-temporal predictions. This approach has been successfully applied in various mechanical fields such as computational fluid dynamics (CFD) (Gao, Deo, & Jaiman, 2024; Li, Robert, Faisal, & Piggott, 2024; Pegolotti et al., 2024), weather forecasting (Lam et al., 2023), cloth simulation (Grigorev, Black, & Hilliges, 2023; Liao, Wang, & Komura, 2024), and solid mechanics (Dalton, Husmeier, & Gao, 2023; Wang, Wang et al., 2024).

Despite the promising capabilities of MGNs, a significant issue can arise during ball drop impact simulations: the prediction of non-physical phenomena where flexible bodies (e.g., the ball and multi-layered display panels in our case) penetrate each other. This penetration accumulates errors over time, leading to the degradation of long-term prediction accuracy and compromising the model's reliability for simulating impacts. To resolve this issue, we propose a physics-constrained MGN that mitigates such non-physical penetration between flexible bodies at each incremental time step during these simulations. First, since our proposed model is based on the conventional MGNs, it represents different mesh geometries as graphs and employs an encoder-processor-decoder (EPD) architecture to effectively capture the complex interactions between two flexible bodies and the dynamic behavior of irregular meshes. The distinguishing feature of our approach is the integration of domain-specific constraints into the loss function. This is achieved by identifying critical boundary nodes, approximating boundary lines using polynomial fitting, and calculating the penetration depth between objects — the degree of the penetration is then served as the additional loss function, which prevents the penetration of two objections. It is expected that this methodology will guarantee that the model adheres to the fundamental physical principles of the simulation, thereby effectively mitigating issues pertaining to object penetration.

To demonstrate the effectiveness of our proposed model, we first analyze the performance and inherent limitations in term of object penetration of the vanilla MGN, underscoring the need for a physics-constrained approach. A parametric study is then conducted to validate the accuracy of the physics-constrained approach in simulating the drop impact of a ball on a multi-layered display panel. This is followed by a detailed comparative evaluation between the vanilla MGN and the physics-constrained MGN, with emphasis on computational cost and predictive accuracy in complex drop impact scenarios. The robustness of our model is further showcased through a case study involving noise injection at varying magnitudes and components, aimed at mitigating error accumulation across temporal predictions. In addition, to enhance model stability and reliability in predicting subsequent node positions over time during the inference, we compare two approaches: predicting

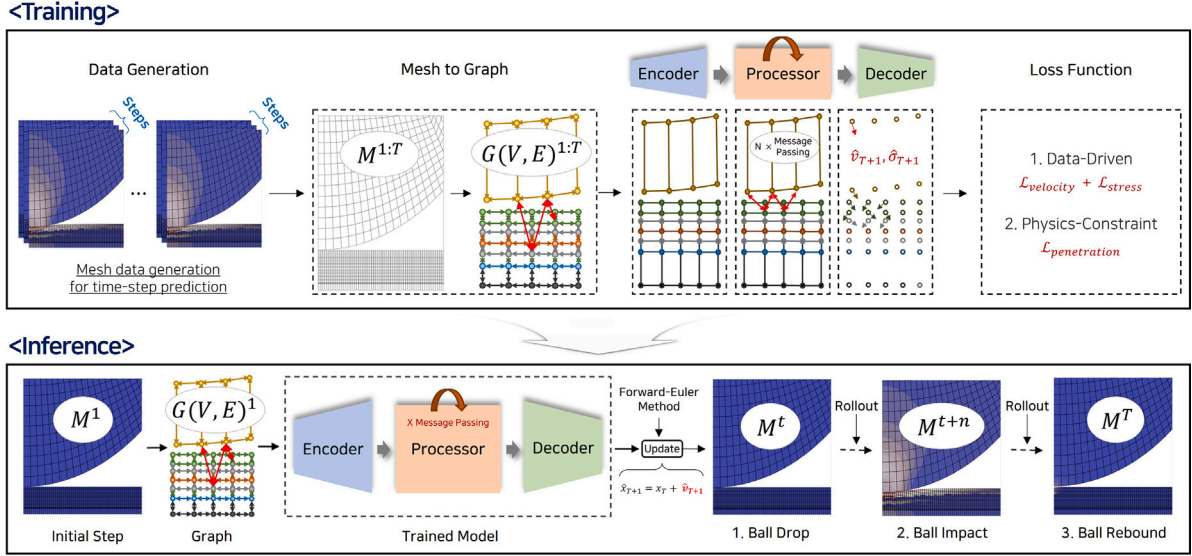


Fig. 1. Two-phase flowchart illustrating the training and inference processes of the physics-constrained MGN.

relative changes (displacement or velocity) between current and future states, and directly predicting absolute node positions. Finally, we evaluate the model's generalization capabilities by examining its performance in extrapolating beyond the initial range of design variables. To further demonstrate the model's practical utility, we couple the proposed predictive model with a gradient-free optimization algorithm, creating a real-time emulator to find optimal configurations of multi-layered display panels. Specifically, we optimize the design variables (thickness of the optically clear adhesive (OCA) layers between multi-layer display panels in our study) to minimize stress distribution within light-emitting materials, leveraging our pretrained predictive models. Our results demonstrate that the proposed model maintains higher accuracy in optimization outcomes compared to a vanilla model when validated against simulation data.

The main contributions of this study can be summarized as follows:

1. We develop a physics-constrained MGN model, which integrates domain-specific physical loss functions to mitigate non-physical penetration phenomena during drop impact tests between two flexible bodies.
2. We further improve the performance of the physics-constrained MGN by determining the optimal noise magnitude to inject into the training dataset and applying this noise across all components involved in the drop impact test. This approach improves the model in terms of error accumulation, thereby ensuring its robust predictive accuracy over long-term predictions.
3. Our analysis on model stability in spatio-temporal predictions demonstrates that inferring future node positions by predicting relative changes (e.g., displacement or velocity) between current and future states consistently outperforms direct absolute position predictions.
4. To evaluate the model's generalization performance, we test the model beyond the training range with respect to the design variables, successfully verifying the reliable performance even in the extrapolated conditions.
5. We apply our proposed model to optimize the OCA thickness of multi-layered display panels, demonstrating its effectiveness in design optimization. By achieving more accurate predictions of maximum stress and deformation, the physics-constrained MGN again proves its superior performance over the vanilla MGN.

Table 1

Mesh M to graph representation $G(V, E^M, E^C)$.

Mesh structure M	Graph representation $G(V, E^M, E^C)$
$C_{quad}, x^m, x^w, n_i, r_w$	$\rightarrow V : \{\phi(n_i), x_i^w, \dot{x}_i^w\}, E^M : \{u_{ij}, u_{ij} , x_{ij}^w, x_{ij}^w \}, E^C : \{x_{ij}^c, x_{ij}^c \}$

2. Physics-constrained MeshGraphNet

Our framework for predicting the nonlinear dynamics of multi-layered display panels under drop impact integrates graph-based learning with physics constraints. As depicted in Fig. 1, the framework transforms mesh data M into a graph $G(V, E)$. This graph comprises a set of nodes V and a set of edges E , which represent the connections between adjacent nodes. Building on this graph structure, our model employs the encoder-processor-decoder (EPD) architecture proposed by Sanchez-Gonzalez et al. (2020), facilitating effective learning and inference in drop impact simulations. In this study, we specifically tailor the predictive model to enhance performance through embedding a penetration physics constraint into the loss function. This constraint is included as a penalty term, guiding the model to produce outputs consistent with physical laws. As a result, integrating this physics constraint aims to improve the accuracy, stability, and generalization of the model's predictions, ensuring they align with the underlying physics of the problem.

This section proceeds as follows: Section 2.1 outlines the steps involved in transforming mesh structures into graph representations. Section 2.2 provides an in-depth analysis of the proposed encoder-processor-decoder architecture, emphasizing its components and their roles in capturing complex system dynamics. Section 2.3 discusses the formulation of the physics-constrained loss function, with a focus on integrating physical constraints of penetration as penalty terms.

2.1. Transforming mesh to graph structure

The process of transforming the mesh structure M into a graph representation $G(V, E^M, E^C)$ is detailed in Algorithm 1, with components summarized in Table 1.

The mesh M comprises quadrilateral cell indices C_{quad} , mesh coordinates x^m , world coordinates x^w , and node types n_i . The mesh coordinates x^m represent the intrinsic positions within the mesh, preserving

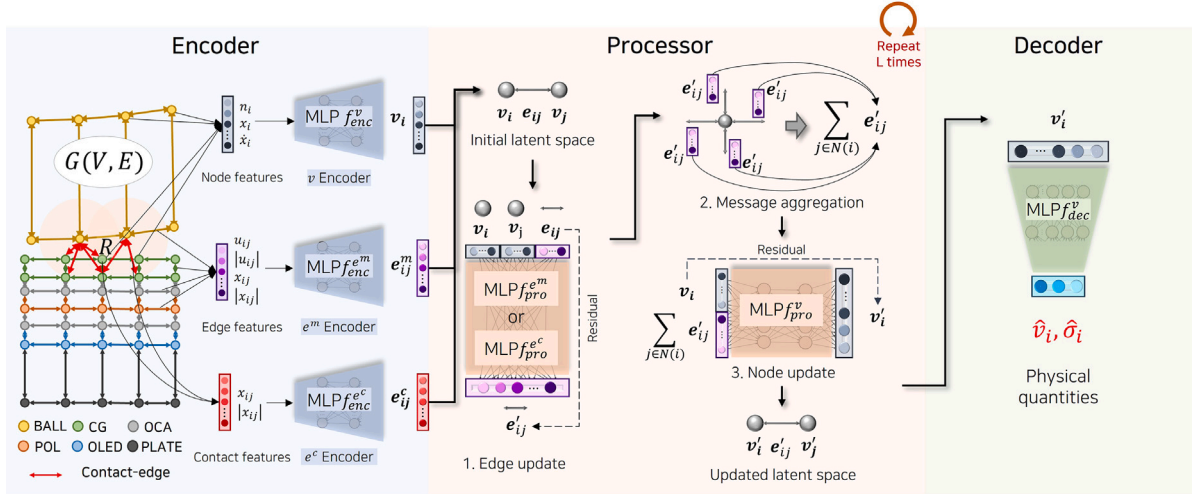


Fig. 2. Specialized MGN architecture for multi-layered display panel impact simulation.

the internal structure over time. In contrast, the world coordinates x_i^w define the actual positions of nodes in the simulation space, which is crucial for modeling dynamics and interactions with the external environment. The node types n_i specify the function of each node i , which can represent entities such as ball, display panel components, or boundary conditions. A contact radius r_w is also defined to establish contact edges between nearby nodes that are not directly connected in the mesh but can influence each other.

The converted graph structure is as follows: Node features v_i are constructed by concatenating the one-hot encoded node type $\varphi(n_i)$, the world space position x_i^w , and the world space velocity $\dot{x}_i^w$. Mesh edge features e_{ij}^m represent the connectivity and deformation of the mesh. These include the relative position in mesh space $u_{ij} = x_i^m - x_j^m$ and its magnitude $|u_{ij}|$, along with the relative position in world space $x_{ij}^w = x_i^w - x_j^w$ and its corresponding magnitude $|x_{ij}^w|$. Contact edge features e_{ij}^c used for interactions between nodes within the contact radius r_w but not directly connected, include the relative position $x_{ij}^w = x_i^w - x_j^w$ and its magnitude $|x_{ij}^w|$.

This graph representation models complex dynamics through two key components: mesh edges E^M that preserve structural integrity through fixed connectivity, and contact edges E^C that enable modeling of dynamic non-local interactions between spatially proximate objects such as collisions or contact forces with other objects in the simulation. To fully utilize this rich graph structure, we employ an EPD architecture, detailed in Section 2.2.

2.2. Encoder-processor-decoder architecture

As the foundation of our approach, we leverage the encoder-processor-decoder (EPD) architecture, as proposed by Sanchez-Gonzalez et al. (2020), to model drop impact simulations, as illustrated in Fig. 2. A comprehensive list of the materials used for the ball and multi-layer display panels can be found in Table 2. This architecture comprises three primary components: the encoder, the processor, and the decoder.

Encoder structure The encoder serves as the initial stage of our model, transforming the input graph structure $G(V, E^M, E^C)$ into a high-dimensional latent space. This transformation is crucial for capturing the system's topological features and initial conditions. To perform this mapping, we utilize a multilayer perceptron (MLP), which enables us to learn rich representations of both nodes and edges. Specifically,

Table 2

Material properties of the dataset.

Layer	Elastic modulus [GPa]	Poisson's ratio	Thickness [μm]
Ball	200	0.3	5000 (radius)
Cover glass (CG)	77	0.21	100
Optically clear adhesive 1 (OCA ₁)	0.01	0.45	50~150
Polarizer (POL)	4	0.33	50
Optically clear adhesive 2 (OCA ₂)	0.01	0.45	50~150
organic light emitting diodes (OLED)	5.15	0.3	30
Aluminum plate (PLATE)	68.9	0.33	1200

the node features v_i are derived through the application of this neural network, followed by a non-linear activation function and layer normalization (LN) (Ba, Kiros, & Hinton, 2016). The encoded node representation v_i is given by:

$$v_i = f_{enc}^v(v_i) = \text{LN} \left(\sigma \left(W^{(k)} v_i^{(k-1)} + b^{(k)} \right) \right) \quad (1)$$

Similarly, the edge features, denoted as e_{ij}^m and e_{ij}^c for different types of edges, are encoded using the same mechanism. The encoded edge representations are expressed as:

$$e_{ij}^m = f_{enc}^m(e_{ij}^m) = \text{LN} \left(\sigma \left(W^{(k)} e_{ij}^{m(k-1)} + b^{(k)} \right) \right) \quad (2)$$

$$e_{ij}^c = f_{enc}^c(e_{ij}^c) = \text{LN} \left(\sigma \left(W^{(k)} e_{ij}^{c(k-1)} + b^{(k)} \right) \right) \quad (3)$$

where f_{enc}^m and f_{enc}^c are encoding functions implemented as networks that take e_{ij}^m and e_{ij}^c , respectively. Here, $W^{(k)}$ and $b^{(k)}$ denote the weight matrix and bias vector of the k th layer, respectively. σ represents the ReLU activation function. LN refers to layer normalization, which is essential in this context as it stabilizes and accelerates the training process by normalizing the inputs across the features for each data point.

$$\text{LN}(x) = \frac{x - \text{E}[x]}{\sqrt{\text{Var}[x] + \epsilon}} \gamma + \beta \quad (4)$$

where $\text{E}[x]$ and $\text{Var}[x]$ represent the mean and variance of the input features, respectively. ϵ is a small constant added for numerical stability, and γ and β are learnable parameters that allow the transformation to be rescaled and shifted. By incorporating LN and non-linear activation functions into the MLPs used in our architecture as well as the encoder, we can effectively capture the complex relationships within the graph

structure, leading to more robust and generalizable representations of the data.

Processor structure The processor models the complex interactions and dynamics between nodes in the graph. It utilizes a message-passing scheme in which each node receives and processes messages (i.e., latent vectors) from its adjacent nodes connected by various edges (e.g., mesh edges, contact edges). This process is implemented using GNNs, which update the latent representations of nodes based on the information received.

As illustrated in Fig. 2, the message-passing scheme within the processor can be divided into three primary steps:

1. **Edge update.** The edge update is represented by Eq. (5). For each edge (i, j) , the updated edge e'_{ij} is computed as:

$$e'_{ij} = f_{pro}^e(v_i, v_j, e_{ij}) = \text{LN} \left(\sigma \left(W^{(l)} \begin{bmatrix} v_i \\ v_j \\ e_{ij} \end{bmatrix}^{(l-1)} + b^{(l)} \right) \right) \quad (5)$$

Here, f_{pro}^e is the edge update function, which takes as input the features of nodes v_i and v_j , and the current edge feature e_{ij} . As a result, the updated edges can allow the model to characterize the relationship between connected nodes.

2. **Message aggregation.** In Eq. (6), for each node v_i , the updated edges connected to its neighboring nodes v_j are aggregated:

$$\sum_{j \in N(i)} e'_{ij} \quad (6)$$

This sum operation collects all the messages passed to node i from its neighbors, preparing the information for the node update step.

3. **Node update.** Finally, the node update is shown as Eq. (7):

$$v'_i = f_{pro}^v \left(v_i, \sum_{j \in N(i)} e'_{ij} \right) = \text{LN} \left(\sigma \left(W^{(l)} \begin{bmatrix} v_i \\ \sum_{j \in N(i)} e'_{ij} \end{bmatrix}^{(l-1)} + b^{(l)} \right) \right) \quad (7)$$

Here, f_{pro}^v is the node update function. It takes as input the current node feature v_i and the aggregated messages from its adjacent nodes.

The processor can consist of multiple message-passing layers (denoted by l in the equations), allowing higher-order interactions and long-range dependencies between the nodes.

Decoder structure The decoder takes the latent representations of nodes, updated by the processor, and maps them back to the output space. Using a series of MLPs that act in the opposite way to the encoder, it predicts desired output quantities such as velocity $\hat{v}_i$ and stress $\hat{\sigma}_i$ for the subsequent time step. The decoder function can be expressed as:

$$\hat{v}_i, \hat{\sigma}_i = f_{dec}^v(v'_i) = \text{LN} \left(\sigma \left(W^{(l)} v'^{(l-1)}_i + b^{(l)} \right) \right), \quad \text{except for LN of last layer } l \quad (8)$$

The final layer omits LN to allow for unbounded output.

2.3. Physical constraint based loss function

In this section, we introduce a physics-constrained loss function designed to improve the accuracy of predictive performance in drop impact scenarios (Fig. 3). This approach specifically addresses the critical issue of unrealistic inter-object penetration, which not only violates fundamental principles of solid mechanics but also leads to inaccurate predictions. As a result, our approach can overcome the limitations of the data-driven spatio-temporal surrogate model proposed by Pfaff et al. (2020), specifically specializing it for the drop impact scenarios and ensuring simulation validity.

To describe our physics-constrained loss function, we present a five-step process:

Step 1. Critical boundary node identification For each step in our trajectory data, which consists of 98 time steps, we identify two critical boundary lines:

$$\begin{aligned} \bullet \text{ Ball bottom boundary: } L_{BALL(t)} &= \left\{ (x, y) \in \mathbb{R}^2 : y \right. \\ &= \min_{i \in N_{BALL(t)} \atop x_i = x} y_i \left. \right\} \\ \bullet \text{ CG top boundary: } L_{CG(t)} &= \left\{ (x, y) \in \mathbb{R}^2 : y = \max_{i \in N_{CG(t)} \atop x_i = x} y_i \right\} \end{aligned}$$

where $N_{BALL(t)}$ and $N_{CG(t)}$ are the sets of nodes representing the ball and CG at time step t , respectively. In this formulation, for each x coordinate, we extract the corresponding y coordinate that represents the minimum (for the ball) or maximum (for the CG) value across all nodes with that specific x coordinate. This ensures that we accurately identify the critical boundary nodes for each boundary line.

Step 2. Polynomial line approximation To efficiently represent the complex shapes of $L_{BALL(t)}$ and $L_{CG(t)}$ and accurately calculate their penetration depth, we approximate each boundary at time step t using n th-degree polynomial functions $P_t(x)$:

$$P_t(x) = \sum_{k=0}^n a_k(t) x^k \quad (9)$$

The choice of polynomial degree is critical, as it directly influences how well the boundary shapes are captured. Lower-degree polynomials may result in underfitting, failing to capture the complexity of the boundaries, while higher-degree polynomials might be too flexible, leading to overfitting. To achieve a high accuracy, we must carefully determine the optimal degree through parameter exploration. This parametric study, which examines the impact of different polynomial degrees on simulation accuracy, will be presented in Section 4.

The coefficients $a(t) = (a_0(t), a_1(t), \dots, a_n(t))^T$ are determined using least squares regression:

$$a(t) = (V^T V)^{-1} V^T y(t) \quad (10)$$

$$V = \begin{pmatrix} 1 & x_1 & \dots & x_1^n \\ 1 & x_2 & \dots & x_2^n \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_m & \dots & x_m^n \end{pmatrix} \quad (11)$$

where V is the Vandermonde matrix and $y(t)$ are the observed y -coordinates at time step t . The Vandermonde matrix V is structured with each row representing powers of x from 0 to n , facilitating polynomial fitting through least squares regression. This regression method is chosen for its ability to minimize the sum of squared residuals, providing a smooth approximation that captures the overall shape of the boundaries while reducing the impact of potential noise in the node positions. With these polynomial approximations $P_t(x)$ for both $L_{BALL(t)}$ and $L_{CG(t)}$, we can now accurately calculate the penetration depth between the ball and the CG at each time step, which is crucial for our penetration loss calculation.

Step 3. Penetration depth calculation The penetration depth is calculated by the difference between the y -values of the ball and CG boundaries. Let $y_{BALL(i,t)}$ and $y_{CG(i,t)}$ denote the y -values of the polynomial functions $P_t(x)$ at the i th node for the ball and CG lines at time step t , respectively. The penetration depth $d(i, t)$ at a given i th node and time step t is given by:

$$d(i, t) = y_{CG(i,t)} - y_{BALL(i,t)} \quad (12)$$

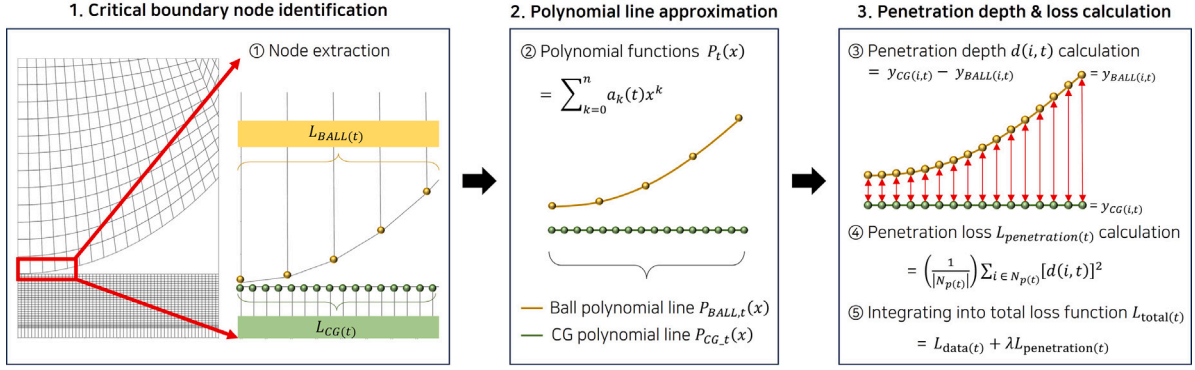


Fig. 3. Flowchart illustrating the incorporation of physical constraints in our model to mitigate inter-object penetration.

If $d(i, t)$ has positive values, it indicates the inter-object penetration, where the ball's surface has penetrated the CG's surface at that specific time step. Negative values indicate there is no contact between the ball and the CG.

Step 4. Penetration loss function We introduce a penetration loss term $L_{penetration}$ based on the mean squared error (MSE) of positive penetration depths. For the time step t , its penetration loss is calculated as:

$$L_{penetration(t)} = \left(\frac{1}{|N_{p(t)}|} \right) \sum_{i \in N_{p(t)}} [d(i, t)]^2 \quad (13)$$

where $N_{p(t)}$ is the set of nodes with positive penetration at time step t , i.e., $N_{p(t)} = \{i \in N(t) | d(i, t) > 0\}$, and $|N_{p(t)}|$ is the number of such nodes. If there are no positive penetrations at time step t , we set $L_{penetration(t)} = 0$. The minimization of such a loss term is expected to hinder the degree of penetration since $d(i, t)$ is expected to decrease.

Step 5. Integration into total loss function The final loss function $L_{total(t)}$ is a weighted sum of the data-driven loss $L_{data(t)}$ (i.e., the sum of velocity and stress loss) and the penetration loss in a single time step t :

$$L_{total(t)} = L_{data(t)} + \lambda \cdot L_{penetration(t)} \quad (14)$$

where $\lambda > 0$ is a hyperparameter controlling the contribution of the penetration loss. We consider the relative scale of the loss weights by comparing the data-driven and penetration losses. Based on this comparison, we set λ such that the magnitude of the penetration loss remains within a similar range as the data loss to ensure balanced optimization. In Section 4, we also present the results of a parametric study on λ , examining several case studies to determine the optimal value that balances penetration prevention and overall simulation accuracy.

3. Data generation and model training

3.1. Data generation

The dataset used in this study is structured as shown in Fig. 4(a) and the properties of each material are listed in Table 2 (Incorporated, 2020; Ma, Jo, Kim, & Kim, 2020; Wu et al., 2022). In this study, the material model is assumed that the OCA behaves as a linear material, rather than using a hyperelastic material model. Based on this assumption, the dataset is generated to include different mesh configurations with different thickness of the OCA, while the initial location of the ball above the plate is fixed at 10 μm . The reason for this height is to alleviate computational burden by removing the unnecessary time

steps to be solved by the computer simulation — therefore the impact velocity is maintained at 0.62631 m/s for all datasets. The overall process of dataset generation is illustrated in Fig. 4(b). Specifically, the data used for training and evaluation is obtained using the explicit dynamics solver in ANSYS Workbench software (Ansys Inc., 2024).

To efficiently use mesh elements for training with limited GPU memory, we simplified the mesh data from 11,526~14,438 nodes to 2403~3117 nodes, focusing on the region where fluctuations in value occur during the ball's impact on the plate. As shown in Fig. 4, the boundary conditions include a fixed constraint applied to the bottom of the PLATE. Additionally, a symmetric condition was imposed to significantly reduce analysis time — by exploiting the symmetry of the model, only right part of the structure needed to be analyzed.

Under these explicit dynamics conditions, a time interval of 4 μs was used in the experiments for a total of 100 time steps. To ensure comprehensive design space, we selected design points using Latin hypercube sampling when designing the experiment (McKay, Beckman, & Conover, 2000). Then, to consider the balance between accuracy and efficiency in our model when splitting train-test trajectories, we calculated the average performance across five MGN models with varying data size, 20, 50, 80, 100, 150 trajectories for the training data, as shown in Fig. 5. From these results, it is observed the error decreases significantly as the data size increases until 80 trajectories. However, the model still shows higher error fluctuations. Therefore, by considering both the decrease in error and the increase in training time, we determined that 100 trajectories offer a good balance between model accuracy and efficiency, making it the appropriate choice for the training dataset.

3.2. Model settings

3.2.1. Model hyperparameters

The networks used in the encoder, processor, and decoder are implemented with MLPs. Each MLP consists of two hidden layers with ReLU activation functions and maintains a 128-dimensional latent vector. The model is trained using the Adam optimizer with a learning rate of 0.001 and a batch size of 1, for 100 epochs. The number of message-passing layers in the processor is set to 15, which is judged to be sufficient for capturing higher-order interactions between the nodes.

3.2.2. Model inference

During model inference, we can predict one future time step (in a process called 'rollout') with respect to a full-step simulation by utilizing the initial state mesh data from unseen trajectories. In a step-by-step rollout prediction, the forward Euler method is applied to obtain the position of the objects at the subsequent time step, as shown

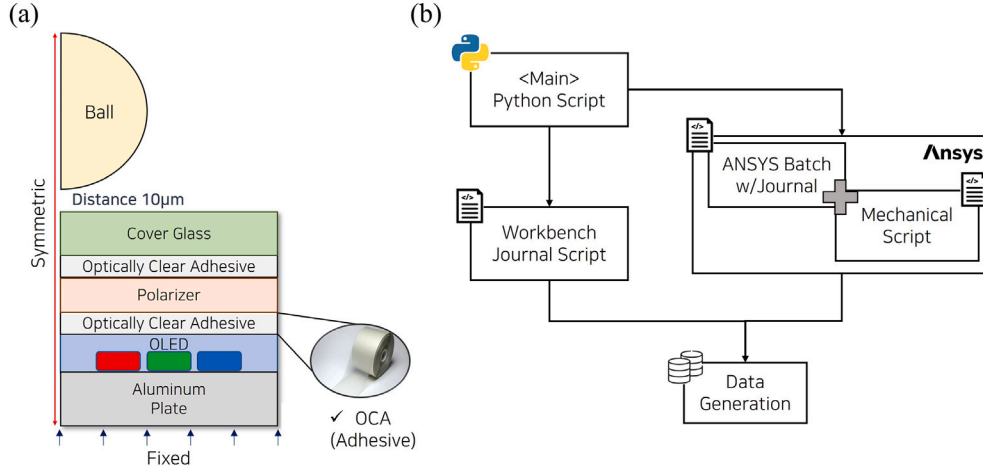


Fig. 4. Schematic of the data generation process: (a) Material structure used in impact test, (b) Automated data generation process.

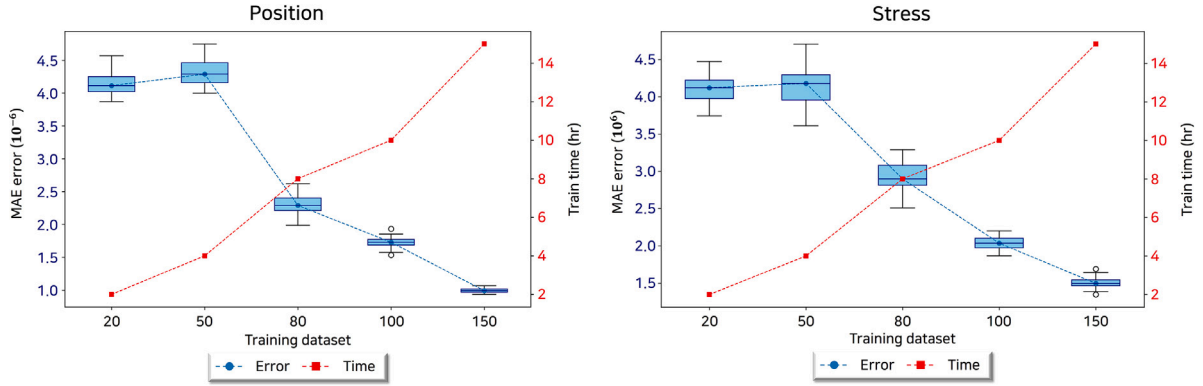


Fig. 5. Validation of model accuracy and efficiency based on the number of training datasets.

in Eq. (15):

$$x_i(t+1) = x_i(t) + \hat{x}_i \Delta t \quad (15)$$

Here, $x_i(t+1)$ represents the position of node i at time step $t+1$, $x_i(t)$ denotes the position of node i at time step t , $\hat{x}_i$ indicates the predicted velocity of node i by the MGN models, and Δt is the time step size, which is fixed as $4 \mu s$ in Section 3.1. In Section 4, we will further explore the impact of using methods other than forward Euler for rollout prediction.

3.2.3. Model implementation and message-passing scheme

Our model is implemented using the PyTorch deep learning framework (Paszke et al., 2019), and the experiments are conducted on an NVIDIA GeForce RTX 3090 GPU with 24 GB of memory. The message-passing layers were implemented using a gather-scatter scheme, employed in PyTorch Geometric (Fey & Lenssen, 2019). This scheme exhibits non-deterministic behavior due to the use of GPU atomic operations, which can lead to varying results across multiple training and testing runs, even when the same random seed is used. While a fully-deterministic implementation is possible by directly utilizing sparse matrix multiplications, it appears to be significantly slower compared to the non-deterministic approach (Gao et al., 2024). Therefore, our study also adopted the non-deterministic characteristic to prioritize computational efficiency.

4. Prediction results

4.1. Limitations of vanilla MGN

MeshGraphNets (MGNs), proposed by Pfaff et al. (2020), offer a versatile graph neural network model applicable to various physical systems. Vanilla MGNs excel in handling unstructured mesh data flexibly and allow for the consideration of interactions between objects through edge connections based on spatial proximity. In this section, we show the performance of vanilla MGN in predicting the nonlinear behavior caused by the drop impact of a ball falling onto multi-layered display panels (Fig. 6). Specifically, we aim to predict the deformations and stresses that occur over time in both flexible bodies.

As Fig. 6, a vanilla MGN fails to maintain physical constraints, resulting in penetration between the ball and the CG layer. It highlights the need for improved approaches for vanilla MGN model that preserve physical consistency while leveraging the computational advantages of GNNs. To address these limitations and improve the accuracy of predictions, in the following section we propose physics-constrained MGN approach specialized for ball drop impact test, and additionally conduct the parametric study to optimize its architecture, ultimately aiming to overcome the drawbacks observed in the vanilla MGN.

The key problems of penetration encountered by vanilla MGN are:

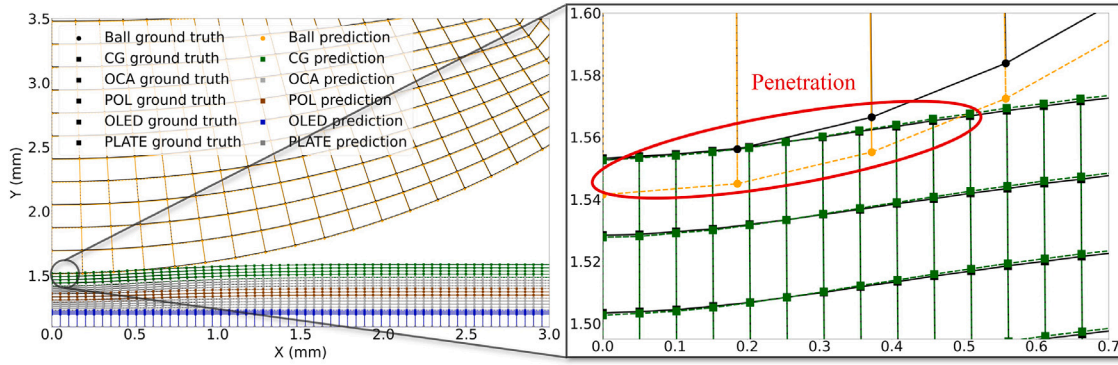


Fig. 6. The prediction performance of a vanilla MGN for ball drop impact on a multi-layered display panel. (Left) Full prediction view. (Right) Zoomed-in view highlighting penetration (red circle) between ball and CG layer. Solid lines/black marker represent ground truth; dashed lines/color markers show predictions.

- **Violation of physical laws:** Impact simulations should satisfy the momentum/energy conservation laws. However, penetration indicates that the surrogate model or contact algorithm fails to enforce these principles correctly. This failure can lead to predictions of post-impact behavior that significantly deviates from physical reality.
- **Inaccurate stress distribution:** Penetration results in the prediction model inaccurately estimating stress distributions due to incorrect representation of contact interactions. This inaccuracy leads to erroneous predictions of object behavior and failure modes, undermining the reliability of the model's outcomes.
- **Cumulative error propagation:** Penetration errors at the initial time steps can propagate through subsequent time steps. This error accumulation leads to progressively divergent predictions, severely undermining the model's long-term predictive capabilities.

4.2. Parametric study to improve the accuracy of physics-constrained model

4.2.1. Exploring polynomial degree used in physics-constrained approach

In Section 2.3, we proposed an approach to prevent penetration phenomena between flexible bodies. To calculate penetration depth accurately, we applied the line approximation technique, approximating both the ball and plate boundaries with polynomial functions. In this section, we further explore how the complexity of boundary shapes influences model performance in this section. Boundary representations are denoted as $B.n_1/P.n_2$, where n_1 and n_2 represent the polynomial degrees used to approximate the boundaries of the ball and display panel geometries, respectively. For example, B.2/P.3 denotes that 2nd order and 3rd order polynomials are used for modeling the shape of ball and display panel, respectively. And the selection of polynomial degrees can be guided by geometric characteristics and expected deformation patterns during impact. Even-degree polynomials (2, 4, and 6) are used for the ball to represent its initial spherical shape and potential deformations during impact. Odd-degree polynomials (3, 5, and 7) are chosen for the panel to account for its initially flat state and to capture the asymmetric bending modes that typically occur during impact. These combinations of ball and display panel approximations are determined to result in a total of 9 cases, which are conducted to assess their impact on model performance: results of these cases are summarized in Table 3.

Our results demonstrate that the decision of polynomial degrees significantly affects the model's capability to precisely predict drop impact behaviors. In this context, MAE refers to the mean absolute error, and RMSE stands for root mean squared error. The B.6/P.5 configuration exhibits superior performance, achieving the lowest errors among all configurations. Compared to the configuration with the

second-lowest errors (B.4/P.5), the B.6/P.5 model shows improvements of 50.6% in position MAE and 63.8% in position RMSE. A key point in this section is that providing higher capacity to the polynomial by increasing its degree does not necessarily enhance performance. For instance, increasing the panel's polynomial degree from 5 to 7 (B.6/P.7) rather leads to a decline in accuracy with position MAE increasing by 76.4% and stress MAE by 31.3%. This suggests an optimal complexity threshold for our approximations, beyond which the model may suffer from overfitting or numerical instability. Computational time remains almost consistent across all configurations, with only a 4.2% variation. This consistency in computational cost allows us to prioritize maximizing accuracy without trade-offs in efficiency when selecting polynomial degrees.

4.2.2. The effect of penetration loss weight on physics constraints

We examine the influence of penetration loss weight (λ) on the model performance as summarized in Table 4. When λ is set to 10, the model achieves optimal results across all metrics, yielding the lowest errors. Compared to the baseline model without penetration loss ($\lambda = \text{None}$), the optimal configuration demonstrates significant improvements: 56.1% reduction in position MAE, 71.8% reduction in position RMSE, 44.0% reduction in stress MAE, and 81.1% reduction in stress RMSE.

However, increasing λ beyond this optimal point does not necessarily enhance performance. When λ is increased to 30, there is an 88.7% increase in position MAE and a 76.3% increase in stress MAE compared to $\lambda = 10$. This suggests that excessively strict penetration constraints can disrupt the balance between the data-driven and physics-constrained loss function, leading to a degradation in overall prediction accuracy. This imbalance may result from the model overemphasizing the physical constraint (penetration) at the expense of capturing the inherent global dynamics of the two objects, which ultimately compromises the global accuracy of the impact test.

For the detailed investigation on the rollout performance of the trained models, we show the mean and standard deviation of the MAE values at each time step for the 50 test trajectories in Fig. 7. As mentioned in Eq. (15), the model makes step-by-step predictions in the inference stage, which can lead to error accumulation over time. According to Fig. 7, the configuration with $\lambda = 10$ demonstrates the most stable performance for both position and stress predictions across the all time steps. Specifically, for position predictions, the configuration with $\lambda = 10$ exhibits consistently lower overall error accumulation. Although stress prediction errors are comparable across all λ values in the initial stages, $\lambda = 10$ shows superior long-term performance. This stability is evidenced by the consistently lower MAE and narrower standard deviation band throughout the simulation. In

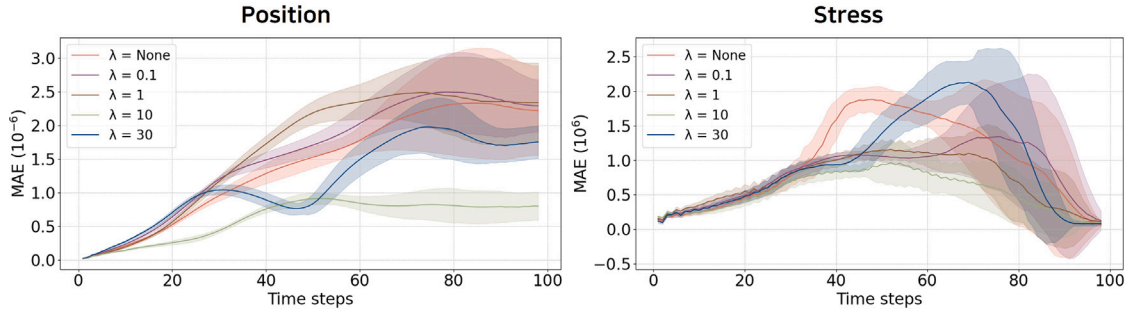


Fig. 7. For different penetration loss weights (λ), the mean and standard deviation of the MAE for position (left) and stress (right) predictions at each time-step rollout, based on all test trajectories.

Table 3

Model performance across various polynomial degree combinations of flexible body boundaries in drop impact prediction is evaluated.

	Polynomial degree								
	B.2/P.3	B.2/P.5	B.2/P.7	B.4/P.3	B.4/P.5	B.4/P.7	B.6/P.3	B.6/P.5	B.6/P.7
Position MAE [10^{-6}]	1.420	0.539	1.273	0.471	0.626	0.909	0.687	0.309	0.545
Position RMSE [10^{-6}]	5.486	1.131	4.917	1.036	1.953	2.931	1.879	0.707	1.288
Stress MAE [10^6]	1.314	1.111	1.276	1.033	0.606	1.054	0.632	0.547	0.718
Stress RMSE [10^6]	8.664	5.362	8.583	5.188	2.228	8.164	2.945	2.223	2.610
Time [h]	21.917	22.017	21.750	21.133	21.583	21.917	21.767	21.233	21.283

Table 4

Validation of varying penetration loss weights in drop impact prediction.

	Penetration loss weight λ				
	None ¹	0.1	1	10	30
Position MAE [10^{-6}]	0.704	0.783	0.830	0.309	0.583
Position RMSE [10^{-6}]	2.509	2.548	2.384	0.707	1.566
Stress MAE [10^6]	0.9775	0.826	0.717	0.547	0.964
Stress RMSE [10^6]	11.744	4.188	2.773	2.223	4.917
Time [h]	19.617	21.617	21.283	21.233	21.9

contrast, The other cases show a rapid error accumulation with higher variability than $\lambda = 10$ case. This suggests that $\lambda = 10$ is the optimal setting for balancing accuracy and stability in long-term predictions.

Fig. 8 illustrates how the model predicts positions at the moment of ball rebound across various λ values. When $\lambda = 0.1$, penetration between the ball and the CG layer is reduced compared to $\lambda = \text{None}$, but both objects rebound later than the ground truth due to insufficient penetration constraint on the system's dynamics. At $\lambda = 1$, the model's predictions remain misaligned, with the ball rebounding too late and the panels slightly early, indicating an imbalance in the constraints. With $\lambda = 30$, penetration is quite mitigated, but both objects are predicted to rebound faster than the ground truth, leading to a loss of accuracy. In contrast, with $\lambda = 10$, no penetration occurs, and the accuracy closely aligns with the ground truth. This configuration achieves a desirable balance between minimizing local penetration and preserving global dynamics, making it the most suitable choice for this task. Therefore, we conclude that $\lambda = 10$ offers the best compromise between preventing penetration and maintaining overall prediction accuracy.

4.3. Comprehensive comparison between vanilla MGN and physics-constrained MGN in predicting impact behavior

This section presents the performance comparison between the vanilla MGN and our physics-constrained MGN models in predicting the behavior of multi-layered display panels subjected to ball drop

impact. First, we assess the statistical performance of two models, the vanilla MGN and the physics-constrained MGN, by analyzing their MAE distributions for position and stress predictions across four experiments per model, each trained under the same settings but with different random seeds, as presented in Fig. 9. Using domain-specific constraints, the physics-constrained model demonstrates significantly improved accuracy and reliability in both position and stress estimations compared to the vanilla MGN. Specifically, the position MAE is reduced by 51.1% in mean, with a 27.8% decrease in standard deviation, while the stress MAE achieves a 44.2% mean reduction, accompanied by a 53.8% decrease in standard deviation. These results underscore the effectiveness and robustness of the model, with notably reduced variability in predictions.

To further clarify the dynamics of these impact systems, we summarize the behavior of two flexible bodies across various time phases.

- Phase 1: Ball drop, no contact
- Phase 2: Ball drop, contact
- Phase 3: Ball rebound, contact
- Phase 4: Ball rebound, no contact

Fig. 10 demonstrates the error accumulation in accordance with step-by-step prediction. Overall, while both models exhibit increasing errors over time, the physics-constrained MGN demonstrates significantly lower error accumulation rates and reduced variability compared to the vanilla MGN. Specifically, the vanilla MGN struggles to accurately model the position of objects due to penetration by drop impact. This results in a noticeable divergence and rapidly increasing MAE, with continuous error accumulation, particularly during the rebound phase. Consequently, the vanilla MGN suffers from extreme stress predictions with a significantly higher peak and pronounced fluctuations when the maximum contact occurs (between phase 2 and phase 3) due to the inferior performance at accurately modeling the non-penetration dynamics, ultimately failing to restore back to its original position during the rebound phase.

In contrast, the physics-constrained MGN demonstrates superior performance, especially during drop impact, due to its incorporation of a penetration constraint. As indicated in Table 5, the performance gap between the models becomes evident starting from phase 2, where

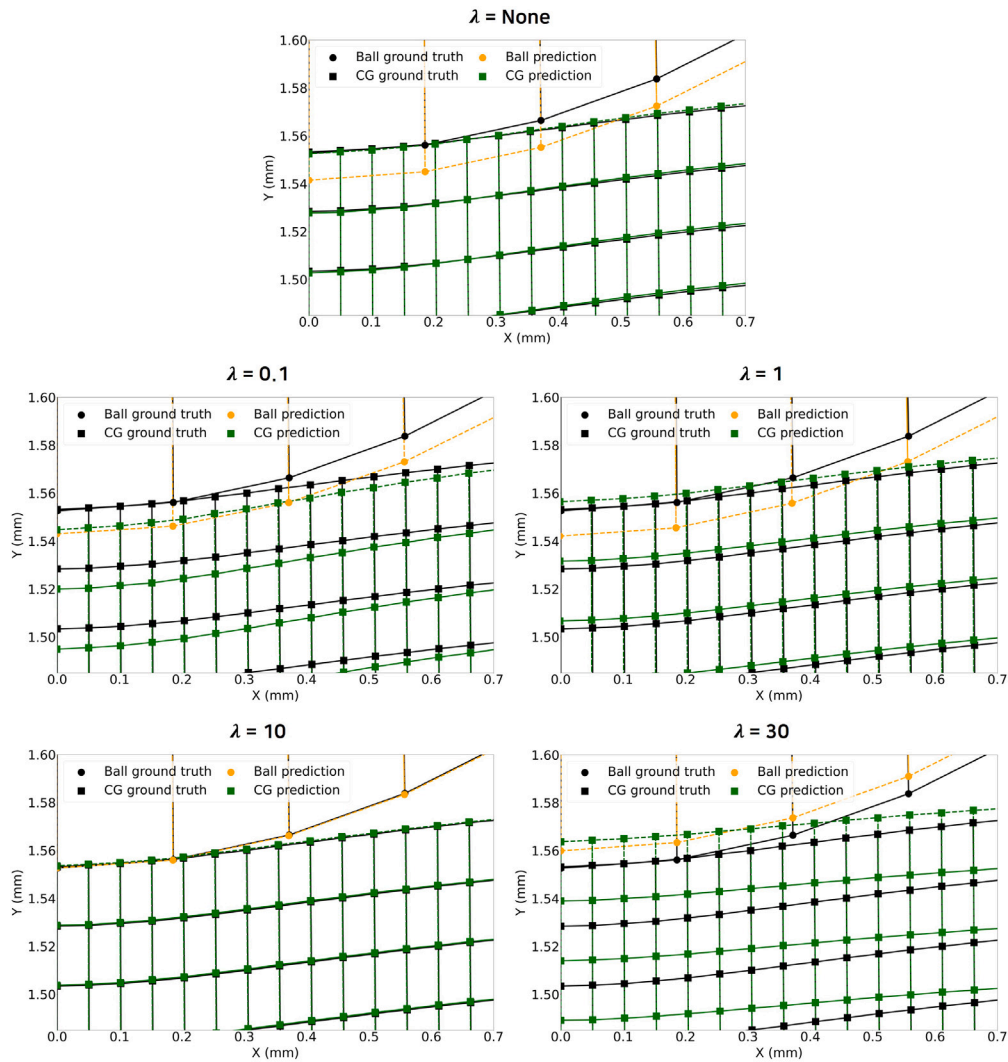


Fig. 8. Visualization of position predictions at the moment of ball rebound for different penetration loss weights (λ). Solid lines/black marker represent ground truth; dashed lines/color markers show predictions.

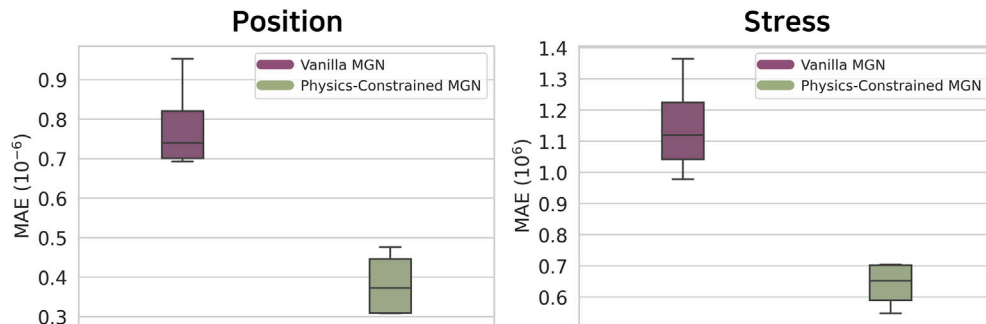


Fig. 9. Statistical validation of the four models, comparing the performance of vanilla MGN and physics-constrained MGN in predicting position and stress under ball drop impact.

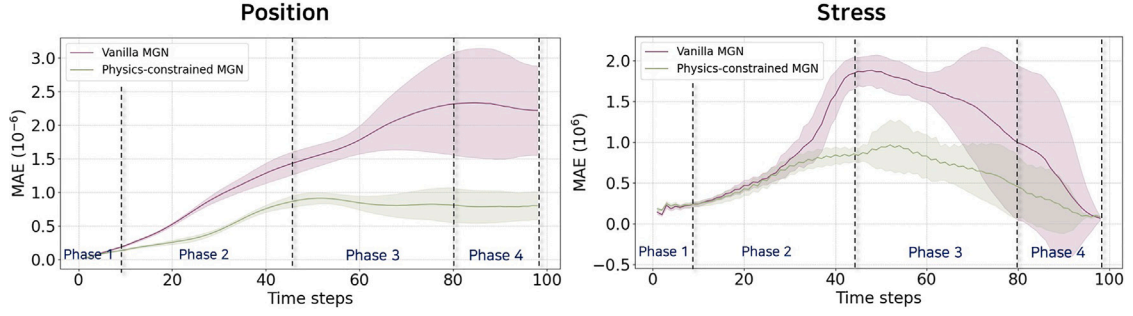


Fig. 10. Rollout-based performance predictions of vanilla MGN and physics-constrained MGN, averaged at each time-step across all test trajectories. This figure represents differences in MAE for both position (left) and stress (right) predictions across various time phases, including drop, impact, and rebound stages.

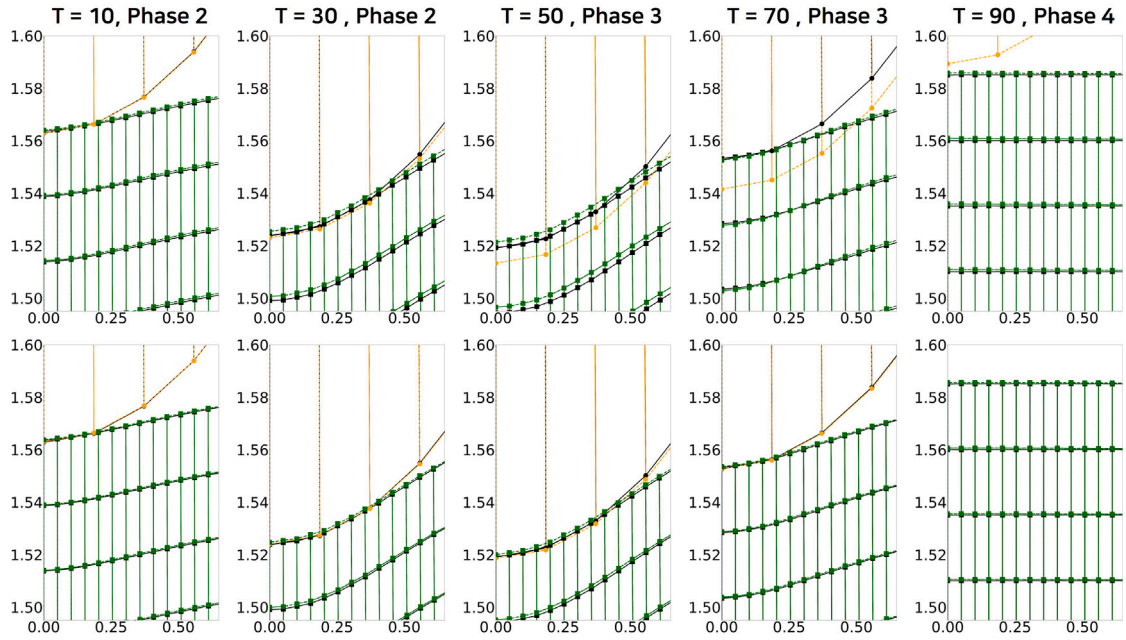


Fig. 11. Visualization for the position prediction at time steps (top: Vanilla MGN; bottom: Physics-constrained MGN; Axes units: 10^{-3}).

Table 5

Phase-wise comparison of MAE for position and stress predictions between vanilla MGN and physics-constrained MGN across different impact phases.

	Model performance comparison during different impact phases							
	Phase 1		Phase 2		Phase 3		Phase 4	
	Position	Stress	Position	Stress	Position	Stress	Position	Stress
Vanilla MGN	0.059	0.174	0.761	0.779	1.892	1.558	2.285	0.513
Physics-constrained MGN	0.056	0.209	0.410	0.553	0.847	0.756	0.792	0.222

the constraint takes effect. Beyond this point, the physics-constrained MGN consistently achieves lower errors in both position and stress predictions, maintaining a low MAE through the contact and rebound phases. Its moderate stress peak during impact and minimal error accumulation in phase 3 highlight its advantage over the vanilla MGN. This performance difference underscores the effectiveness of the physics-constrained model in mitigating penetration issues and ensuring accurate predictions.

We intuitively compare the rollout prediction results of the vanilla MGN and our physics-constrained MGN based on position error (Fig. 11) and stress error (Fig. 12). Initially, at $T = 10$, both models

show comparable accuracy in position prediction. However, as time progresses, significant errors emerge, revealing the strengths of our proposed approach. Specifically, the vanilla MGN shows a gradual decline in accuracy, particularly after $T = 50$, primarily due to penetration issues that result in delayed predictions. This growing discrepancy is clearly illustrated in the zoomed-in view at later time steps. In contrast, physics-constrained MGN consistently maintains a high level of accuracy throughout the entire sequence, with minimal drift from the ground truth. Furthermore, by examining the absolute difference between predicted and ground truth stress $|\sigma - \hat{\sigma}|$ (Fig. 12), we observe that our proposed model achieves substantially lower stress errors

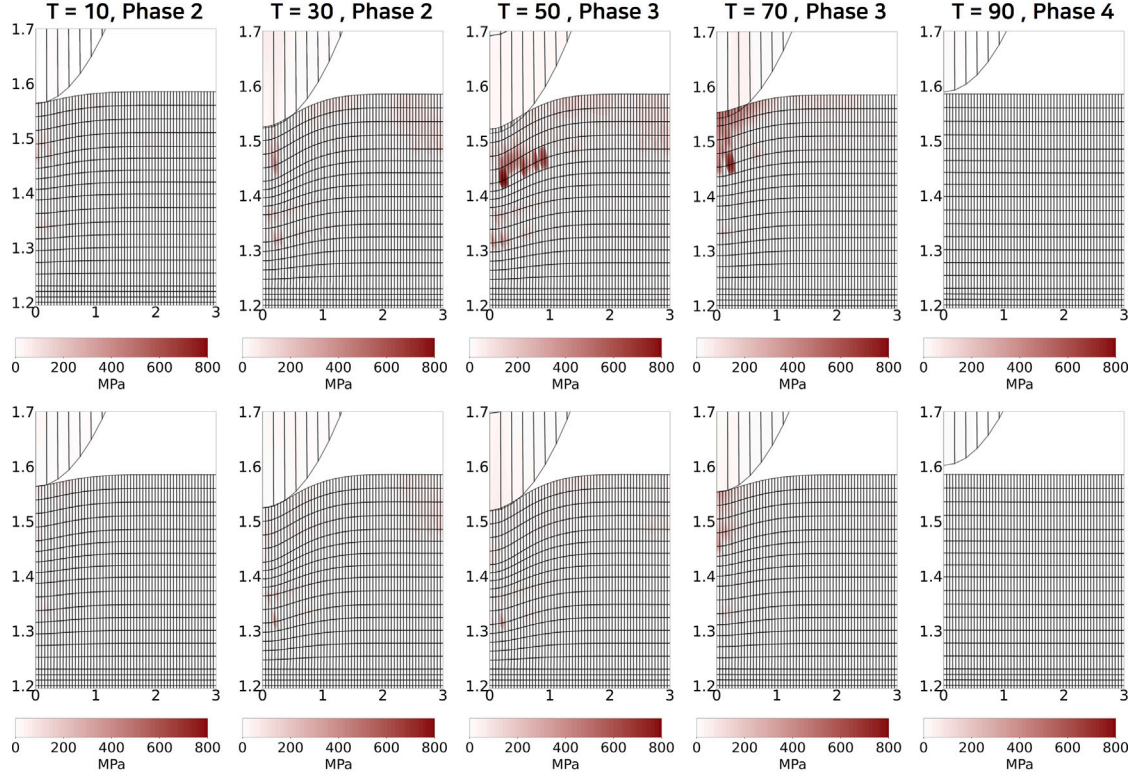


Fig. 12. Visualization of stress error predictions at each time step (top: Vanilla MGN; bottom: Physics-constrained MGN).

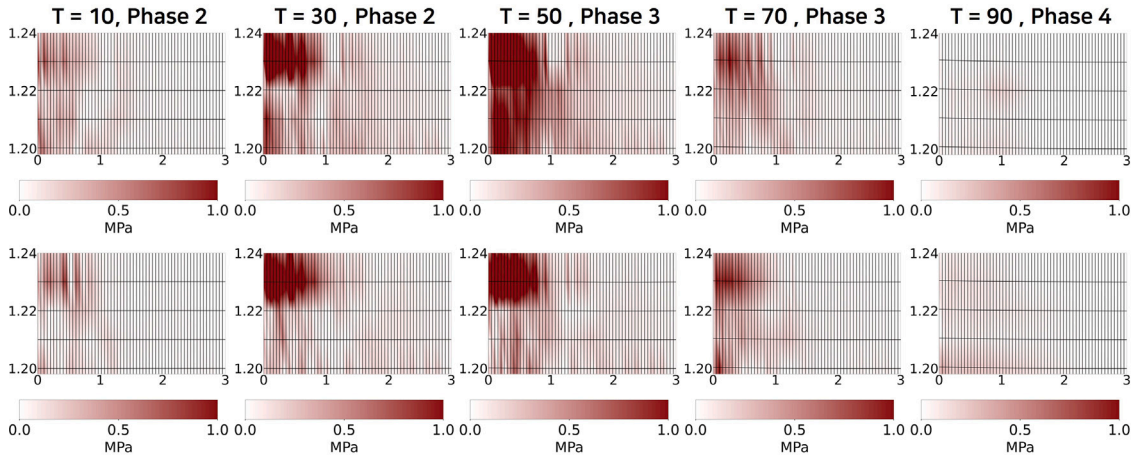


Fig. 13. Visualization of stress error predictions in OLED structure at each time step (top: Vanilla MGN; bottom: Physics-constrained MGN).

across all time steps. In particular, it can be seen that the vanilla MGN model shows a clear trend of increased error at $T = 30, 50$, where the penetration occurs — while the proposed model shows reasonable errors at the same time steps but with relatively much lower stress error than the vanilla one.

Fig. 13 focuses on a stress error distribution over time especially for OLED material, a critical light-emitting component in multi-layered display panels. Physics-constrained MGN still shows superior performance compared to the vanilla MGN model, exhibiting lower stress errors across time steps. This improved accuracy is crucial for optimizing display design, where the thickness of the OCA layer is optimized to

minimize stress on the OLED material. In this regard, accurate stress prediction is essential for maintaining the longevity and performance of OLED displays, as even minor errors can lead to suboptimal designs and reduced durability. The improved precision of our model, reflected in Fig. 13 by the more localized and reduced distribution of the red areas in the lower panels, especially at $T = 50, 70$ where the maximum stress is generated, enables more reliable prediction of the stress distribution prediction. In Section 5, we will delve into the process of optimizing OCA thickness using our high-dimensional prediction model, which builds upon spatio-temporal modeling techniques.

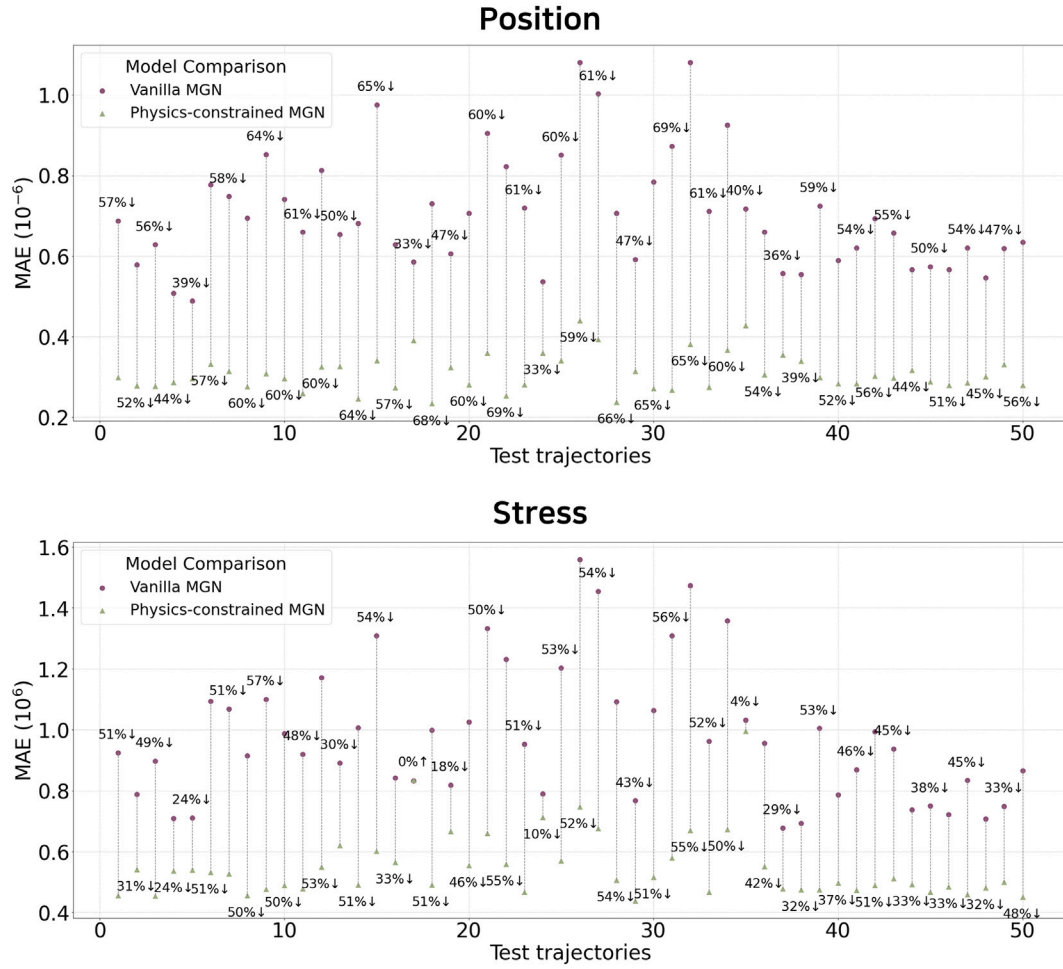


Fig. 14. Comparison of prediction performance between vanilla MGN and physics-constrained MGN per test trajectory. Top: Position prediction errors (MAE across rollout-all). Bottom: Stress prediction errors (MAE across rollout-all). Each percentage indicates the relative decrease in physics-constrained MAE compared to vanilla MAE.

Lastly, the physics-constrained MGN proves that it universally maintains higher performance across all test trajectories, as shown in Fig. 14. The percentages at each trajectory indicate the relative error reduction achieved by our physics-constrained model (green triangular markers) compared to the vanilla MGN (purple circular markers). This consistent improvement in both position and stress predictions emphasizes the remarkable robustness and reliability of our physics-constrained approach throughout all of the test cases, making it a valuable tool for accurate OLED panel design and optimization across diverse scenarios.

4.4. Investigation and enhancement of model robustness and generalization capabilities

4.4.1. Effects of noise injection: A case study across various noise magnitudes and components

In this section, we assess the model's robustness with noise injection approaches based on Gaussian white noise. By applying additive white Gaussian noise to the input data during training, the model's adaptability is expected to be enhanced, enabling it to solve a wide range of complex forward and inverse problems (Sanchez-Gonzalez et al., 2020). This technique introduces small perturbations into the input data (in our case, the global position $x^{(0)}$) used for training, with these perturbations following a Gaussian distribution with a mean of 0 and a standard deviation of σ . Acting as a powerful regularization tool, it helps the model relieve the error accumulation, which can cause

distribution shifts during long-term rollouts (Toshev, Paehler, Panizza, & Adams, 2023; Yang et al., 2024b).

To find the optimal noise magnitudes, we tested four standard deviation values: 0.1, 0.3, 0.7, and 1.5 μm , considering the scale of the training data. As shown in Table 6, while the training time remained relatively stable across different noise injection scenarios (since noise injection process requires insignificant computational cost), the impact of noise magnitude on model performance was significant. Our proposed model showed the best performance with a noise of 0.3 μm , reducing position error by 71.8% and stress error by 81.1% compared to the best vanilla MGN, based on RMSE. Ultimately, this means that by injecting noise levels appropriate to each model, the model can learn to compensate for perturbations that degrade long-term rollout performance through error accumulation.

We also conducted additional experiments to verify the effects of noise when injected into different flexible body components. We consider four cases: None, Ball, Display panels, and All. Table 7 presents the experimental results of the two comparative models based on noise injection for each component case. Both models achieved peak performance when noise was introduced into all system components. This noise-induced enhancement suggests that small perturbations to all flexible body components in impact scenarios can effectively mimic the inherent uncertainties and fluctuations present in object interactions, and therefore reduce the error exacerbated through rollout processes.

Table 6

Investigation of predictive model performance with varying noise magnitudes across all nodes. Note: position errors (MAE, RMSE) are in units of 10^{-6} , while stress errors (MAE, RMSE) are in units of 10^6 .

		Injected noise magnitudes							
		0.1 μm		0.3 μm		0.7 μm		1.5 μm	
		Position	Stress	Position	Stress	Position	Stress	Position	Stress
Vanilla MGN	MAE	3.458	2.588	0.704	0.978	0.915	0.953	1.806	1.551
	RMSE	14.733	16.319	2.509	11.744	4.544	3.914	9.318	9.728
	Time [h]	19.500		19.617		19.617		21.300	
Physics-constrained MGN	MAE	1.203	1.685	0.309	0.547	1.421	1.786	1.504	1.815
	RMSE	4.308	11.766	0.707	2.223	2.446	7.140	4.464	11.824
	Time [h]	21.433		21.233		22.133		22.267	

Table 7

Investigation of predictive model performance with noise injection in each flexible body component. Note: position errors (MAE, RMSE) are in units of 10^{-6} , while stress errors (MAE, RMSE) are in units of 10^6 .

		Body components for noise injection							
		None		Ball		Display panels		All	
		Position	Stress	Position	Stress	Position	Stress	Position	Stress
Vanilla MGN	MAE	2.536	4.431	6.456	4.119	6.685	4.397	0.704	0.978
	RMSE	10.788	26.206	32.351	23.608	32.727	26.705	2.509	11.744
	Time	19.200		19.183		20.400		19.617	
Physics-constrained MGN	MAE	3.826	4.465	3.020	4.747	7.166	5.380	0.309	0.547
	RMSE	18.346	25.339	11.459	28.702	34.108	33.516	0.707	2.223
	Time	21.283		21.283		21.367		21.233	

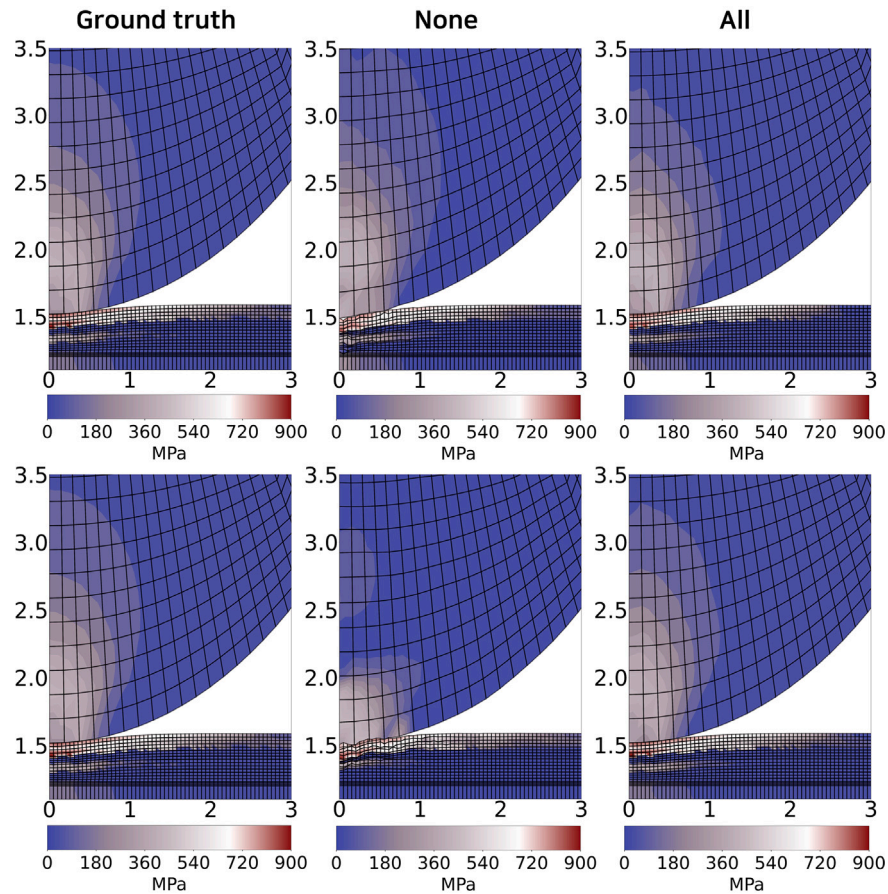


Fig. 15. Stress distribution when maximum stress occurs, showing the effect of noise injection into flexible bodies. Top row: vanilla MGN; Bottom row: physics-constrained MGN. Columns represent ground truth (left), without noise injection (middle), and noise injection into all components (right).

Finally, to provide an intuitive comparison of prediction performance according to the noise injection, we visualized the predicted

behavior and stress distribution when the maximum stress occurs during the inference stage (Fig. 15). It indicates that without the addition

of small perturbations, the model's accuracy is generally lower. In the absence of noise (None case), the model struggles to accurately capture the complex interactions between two objects, particularly through contact nodes and edges: they show unrealistic behavior, such as zigzagging nodes in the upper plate. In contrast, when small perturbations are added to all objects (All case), we observe that the model successfully handles the interactions, producing results very similar to the ground truth for both vanilla and physics-constrained models. Through this, we prove the effectiveness of noise injection in improving the model's ability to generalize and accurately predict complex object interactions in terms of rollout process.

4.4.2. Extrapolation performance: thickness-based predictions beyond training range

In this section, we evaluate the extrapolation capabilities of the models outside the training range with respect to thickness conditions to validate their practical applicability and overall robustness. As shown in Table 2, the OCA thickness in our training dataset ranged from 50 to 150 μm . To assess the models' extrapolation capabilities, we conduct tests under four additional thickness conditions outside this range (Fig. 16). We examine the time-step when maximum stress occurs for each extrapolation thickness condition. When OCA_1 and OCA_2 both have a thickness of 40 μm (first column in Fig. 16), where the thickness of both OCAs is less than the training region so this can be considered the most extreme extrapolation case,² both models have the largest stress error in the OCA_1 plate. This indicates difficulty in accurately predicting complex stress distributions in thin OCA structures. The remarkable decrease in accuracy for thicknesses below the training range suggests limited capability in capturing nonlinear behavior under unseen extreme conditions, indicating that additional approaches or incorporation of training data are required for predictions in these areas. For other thickness conditions (second to fourth column in Fig. 16), vanilla MGN continues to display penetration phenomena and higher errors, particularly in the CG and OCA_1 regions. In contrast, physics-constrained MGN generally avoids penetration and maintains significantly lower stress errors. This demonstrates that incorporating physical laws into the learning process enhances the model's adaptability across a extrapolation conditions.

4.4.3. Performance validation of rollout predictions based on output quantities

As described in Eq. (15), rollout prediction is performed using the forward Euler method. In this section, we evaluate the performance of rollout with different output quantities.

First, we directly predict the absolute node position for the subsequent time step, which simplifies the procedure by eliminating the need for intermediate calculations (Eq. (16)). Second, we predict relative changes, specifically displacement, which enhances the model's sensitivity to capture minor variations effectively (Eq. (17)). Finally, we predict velocity, and then use the forward Euler method to estimate the node position at the next time step.

$$x_i(t+1) = \hat{x}_i(t+1) \quad (16)$$

$$x_i(t+1) = x_i(t) + \Delta \hat{x}_i \quad (17)$$

Here, $x_i(t)$ represents the position of node i at time t , $\Delta \hat{x}_i$ represents the predicted displacement, capturing the difference between the next and current node positions.

² Extrapolating to thinner plates is more challenging than extrapolating to thicker plates due to increased sensitivity to small changes in geometry. Thinner plates experience higher stress concentrations, which magnifies the influence of any inaccuracies in the model's learned behavior. Thicker plates, by contrast, tend to distribute stresses more uniformly, making predictions less sensitive to the extrapolated conditions.

As shown in Table 8, our experiments indicate that direct position prediction leads to significant prediction errors. In contrast, the velocity prediction using the forward Euler method combined with the physics-constrained MGN proves to be the most effective. This approach updates the velocity at each time step and utilizes it to predict the position at the next time step (Eq. (15)), enabling the model to accurately capture continuous motion and provide highly precise predictions for both position and stress, including the fine-scale variations inherent in Lagrangian systems. Additionally, displacement-based predictions show significantly lower errors compared to directly predicting absolute node positions at the next time step, though they still fall short of velocity-based predictions. The superior performance of both velocity-based and displacement-based predictions over direct absolute position predictions highlights the advantages of using relative changes to estimate the next node position.

5. Extension to design optimization

5.1. Problem definition for design optimization of OCA thickness

This section focuses on using the trained MGN models for the design optimization of OCA thickness. To determine the optimal configuration of multi-layered display panels that minimizes stress distribution in the OLED panel, we optimize the thickness of the two OCA layers using a genetic algorithm (GA) with a single objective. As outlined in Eq. (18), the optimization problem is formulated with specific constraints:

$$\begin{aligned} \min_{\mathbf{X}} \quad & \hat{f}(\mathbf{X}) \\ \text{w.r.t } \quad & \mathbf{X} = [T_{\text{OCA}_1}, T_{\text{OCA}_2}] \\ \text{subject to} \quad & 50 \mu\text{m} \leq \mathbf{X} \leq 150 \mu\text{m}, \\ & 150 \mu\text{m} \leq T_{\text{OCA}_1} + T_{\text{OCA}_2} \leq 250 \mu\text{m}, \end{aligned} \quad (18)$$

Here, $\hat{f}(\mathbf{X})$ is calculated as the average stress of nodes where $x < 3$ in the mesh grid: this region is specifically chosen because it represents the area where intensive stress occurs due to impact, making it crucial for optimizing stress distribution.

The optimization process, using surrogate models to accelerate computations, was completed in 2 h and 24 min. We employed a genetic algorithm (GA) with a population size of 100, running for 50 generations, to minimize stress distribution. Given the nature of spatio-temporal prediction, most of the computation time was spent on performing rollouts, which scales with the population and generation size. However, once the initial shape is established, full simulations for various scenarios can be efficiently validated, making this approach significantly more efficient compared to direct numerical simulation.

5.2. Comparison of consistency between ground truth and model predictions

In this section, we compare the consistency between ground truth data and the predictions made by both the vanilla MGN and the physics-constrained MGN, focusing on the acquired optimal solution. The purpose of this comparison is to verify how accurately each model predicts the stress distribution in OLED panels and thus how much validity the optimal solution has compared to the ground truth solver, which in our case is an explicit dynamics solver. Assessing this consistency is crucial for validating the reliability of each model and highlights the significance of achieving accurate optimal results in real-time through surrogate-based stress prediction.

Table 9 summarizes the optimal OCA layer thickness values derived from each predictive model, the mean stress values within OLED panels from both ground truth solver and model predictions, and the percentage error between those stress values. The noteworthy point is that those two MGN models derived two optimal solutions with totally different OCA thickness: vanilla gets optimal design with maximum OCA_2 value whereas physics-constrained model obtains design with maximum OCA_1 value. While the vanilla MGN initially seemed

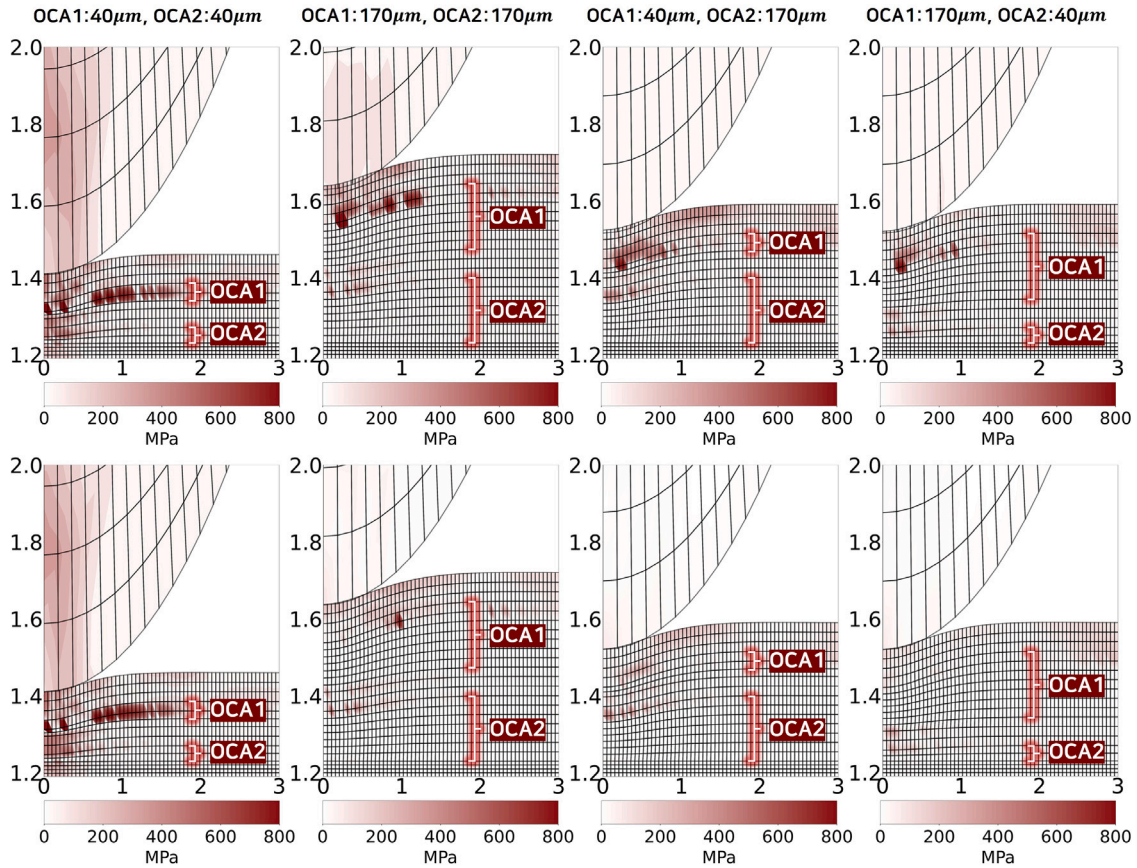


Fig. 16. Stress error distribution for extrapolation cases across OCA thicknesses outside the training range. Top row: vanilla MGN; Bottom row: physics-constrained MGN. OCA₁ and OCA₂ values show thicknesses (in μm) for each extrapolation case. The time steps at which the maximum stress occurs in each case are 30, 49, 43, 43.

Table 8

Comparison of rollout prediction performance between the vanilla MGN model and the physics-constrained MGN model based on three output quantities.

		Position		Displacement		Velocity	
		Position	Stress	Position	Stress	Position	Stress
Vanilla MGN	MAE	1458.494	59.638	1.051	1.130	0.704	0.978
	RMSE	2218.880	85.306	3.148	7.718	2.509	11.744
	Time [h]	19.500		19.433		19.617	
Physics-constrained MGN	MAE	1318.342	34.451	0.485	0.797	0.309	0.547
	RMSE	2070.163	60.919	1.268	3.472	0.707	2.223
	Time [h]	21.383		21.833		21.233	

Table 9

Comparison of optimization results for vanilla and physics-constrained models.

	OCA thickness optimization	
	Vanilla MGN	Physics-constrained MGN
Optimal values (μm)	$T_{\text{OCA}_1} = 100.364$, $T_{\text{OCA}_2} = 149.635$	
Mean stress of ground truth	1.995 MPa	$T_{\text{OCA}_1} = 145.207$, $T_{\text{OCA}_2} = 99.322$ 1.891 MPa
Mean stress of prediction	1.666 MPa	1.860 MPa
Error	16.45%	1.67%

promising with lower stress values, comparison with the ground truth revealed that the physics-constrained MGN produced a significantly more accurate result. Evaluating these optimal designs, the results indicate that the physics-constrained MGN demonstrates superior consistency, reducing the error rate by approximately 89.8%, compared to the vanilla MGN model. Our proposed model significantly improves accuracy in predicting actual stress distribution, thereby providing a more reliable method for optimizing OLED panel design.

6. Conclusions, limitations and future work

While conventional MGNs have shown promise in spatio-temporal prediction tasks, their exclusive reliance on data-driven modeling presents significant challenges in accurately representing physical phenomena such as drop impact tests. In this study, we proposed a physics-constrained MGN to address the challenges of accurately predicting the nonlinear dynamics of multi-layered display panels under

ball drop impact. Specifically, our model incorporates physical constraints directly into the existing loss function by calculating the penetration depth between objects and adding it as a weighted loss term. This approach ensures compliance with the fundamental physical principle of non-penetration during object contact. To demonstrate the effectiveness of our proposed model, we first conducted parametric study demonstrating the importance of polynomial degree selection for object boundary approximations and then explored the effects of the penetration loss weight (λ) that balances accuracy and stability in drop impact scenario. Results indicated that our physics-constrained MGN significantly outperformed the vanilla MGN with reduced error accumulation during rollout predictions owing to its physics-constrained loss function. Moreover, it universally achieved lower errors across all test datasets, effectively mitigating penetration issues also in the unseen impact cases. We additionally validated the model's robustness through various noise injection experiments, adjusting both noise magnitude and target objects to be contaminated by the noise. Introduction of noise across both ball and plate resulted in significantly lower errors than the case without noise, while maintaining consistent spatio-temporal accuracy at each time step, even with previously unseen data. Furthermore, we also demonstrated the effects of rollout predictions based on different physical output quantities, showing that predicting relative changes (such as displacement and velocity) is significantly more stable than directly predicting absolute positions at the next time step. Then, we confirmed the model's extrapolation performance by exploring beyond the training range of design variables. Except for the excessive condition, where the both design variables were thinner than the training range, the physics-constrained model showed the superior extrapolation performance compared to the baseline vanilla model. Finally, we applied our physics-constrained MGN as a spatio-temporal surrogate model in the design optimization procedure for optimizing the thickness of OCA layers to minimize stress distribution in OLED panels. The optimal solution obtained by the proposed model shows better performance than that obtained by the vanilla model, and also the discrepancy between the ground truth (computer solver) and the prediction by the models is found to be much smaller in the physically constrained model compared to the vanilla model.

Here are the primary outcomes of our research:

1. The results demonstrated that selecting the appropriate combination of polynomial degrees for modeling of each object boundary and introducing a penetration loss weight of $\lambda = 10$ led to significant improvements in model performance. Compared to the model without applying polynomial approximations and a penetration loss weight, our model achieved a 71.8% decrease in position RMSE and an 81.1% decrease in stress RMSE.
2. Injecting noise into both ball and plate with a magnitude of $0.3 \mu\text{m}$ yielded the best results, reducing errors by 64.3% in position MAE and 81.1% in stress MAE compared to the case without noise injection.
3. We further found that the rollout with velocity-based prediction achieved a 70.5% reduction in position MAE and a 51.6% reduction in stress MAE compared to direct position prediction. In this regard, we demonstrated that predicting relative changes, such as displacement or velocity, captures local variations and dynamics, which helps reduce cumulative errors during rollouts and is significantly more stable and effective than directly predicting absolute positions.
4. We applied the physics-constrained MGN for the design optimization of OCA layer thickness, achieving an optimal solution with an error rate of 1.67% compared to the ground truth, which represents an approximate 89.8% reduction in error compared to the vanilla MGN's optimal solution. It demonstrates that the

proposed model provides not only a more accurate prediction but also better optimization outcomes for minimizing stress distribution in OLED panels.

The physics-constrained MGN focuses on the specific application of drop impact testing for OLED displays, but there are limitations to consider when extending this approach to new applications compared to the vanilla MGN. This study primarily addresses relatively simple geometries, where boundary modeling was effectively performed using simple polynomial fitting. However, to account for the flexibility required for more complex geometric variations, advanced techniques such as signed distance functions (SDFs) for modeling intricate boundary geometries or B-splines, widely used in CAD for constructing smooth and precise curves, need to be considered. Incorporating these enhancements would not only enable the model to generalize across different fields but also expand its applicability to practical use cases, such as analyzing structural deformations and thermal runaway in battery packs or cells subjected to external impacts, simulating car crash scenarios to evaluate vehicle deformation and energy absorption during collisions, and assessing material deformations in both internal components and external structures during helmet impact absorption tests.

As part of our future work, we first plan to integrate physics-informed neural networks (PINNs) (Bolandi, Sreekumar, Li, Lajnef, & Boddeti, 2023; Yang, Kim et al., 2024) to embed partial differential equations into the loss function for accurate modeling of stress wave propagation and localized deformations. Next, to improve computational efficiency, we will adopt graph pooling methods (Fortunato, Pfaff, Wirsberger, Pritzel, & Battaglia, 2022) for multiscale representation while preserving critical structural features. Finally, we aim to explore hypergraph structures (Gao et al., 2024) to capture higher-order relationships essential for modeling finite element interactions in collision scenarios.

CRediT authorship contribution statement

Jiyong Kim: Conceptualization, Formal analysis, Investigation, Methodology, Visualization, Writing – original draft, Writing – review & editing, Supervision, Software. **Jangseop Park:** Methodology, Software. **Nayong Kim:** Software. **Younyeol Yu:** Software. **Kiseok Chang:** Software. **Chang-Seung Woo:** Software. **Sunwoong Yang:** Methodology, Supervision, Writing – review & editing. **Namwoo Kang:** Funding acquisition, Project administration, Resources, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by LG display, the National Research Foundation of Korea grant (2018R1A5A7025409), the Ministry of Science and ICT of Korea grant (No. 2022-0-00969, No. 2022-0-00986, No. GTL24031-300, and No. RS-2024-00355857), the Ministry of Trade, Industry & Energy grant (RS-2024-00410810).

Appendix. Details of converting mesh structures into graph representations

Algorithm 1 Conversion from mesh to graph structure**Input:** Mesh $M = (C_{\text{quad}}, x^m, x^w, n_i)$, contact radius r_w **Output:** Graph $G(V, E^M, E^C)$, where V is the set of nodes v_i , E^M is the set of mesh edges e_{ij}^m , and E^C is the set of contact edges e_{ij}^c **Node features:**1: $v_i \leftarrow \text{CONCAT}(\varphi(n_i), x_i^w, \dot{x}_i^w)$

▷ Create node features by combining node type, position, and velocity

Edge features:2: Extract edge pairs (i, j) from C_{quad}

▷ Obtain edges from quadrilateral cells

Mesh edge features:3: **for all** mesh edges (i, j) **do**4: $u_{ij} \leftarrow x_i^m - x_j^m$

▷ Relative position in mesh space

5: $x_{ij}^w \leftarrow x_i^w - x_j^w$

▷ Relative position in world space

6: $e_{ij}^m \leftarrow \text{CONCAT}(u_{ij}, |u_{ij}|, x_{ij}^w, |x_{ij}^w|)$

▷ Form mesh edge features

7: $e_{ij}^m \leftarrow \text{NORMALIZE}(e_{ij}^m)$

▷ Normalize features

8: **end for****Contact edge features:**9: Compute distances $D_{ij} = \|x_i^w - x_j^w\|_2^2$ 10: Exclude node pairs where $D_{ij} > r_w^2$, self-connections and existing mesh edges11: **for all** contact edges (i, j) **do**

▷ Relative position in contact space (except for the nodes from lines 10))

12: $x_{ij}^c \leftarrow x_i^w - x_j^w$

▷ Form contact edge features

13: $e_{ij}^c \leftarrow \text{CONCAT}(x_{ij}^c, |x_{ij}^c|)$

▷ Normalize features

14: $e_{ij}^c \leftarrow \text{NORMALIZE}(e_{ij}^c)$ 15: **end for**16: **return** v_i, e_{ij}^m, e_{ij}^c **Data availability**

Data will be made available on request.

References

- Anslys Inc. (2024). ANSYS workbench: Simulation integration platform. URL: <https://www.ansys.com/products/ansys-workbench>. (Accessed 8 September 2024).
- Asher, M. J., Croke, B. F., Jakeman, A. J., & Peeters, L. J. (2015). A review of surrogate models and their application to groundwater modeling. *Water Resources Research*, 51(8), 5957–5973. <http://dx.doi.org/10.1002/2015WR016967>.
- Ba, J. L., Kiros, J. R., & Hinton, G. E. (2016). Layer normalization. <http://dx.doi.org/10.48550/arXiv.1607.06450>, arXiv preprint arXiv:1607.06450.
- Bolandi, H., Sreekumar, G., Li, X., Lajnef, N., & Boddetti, V. N. (2023). Physics informed neural network for dynamic stress prediction. *Applied Intelligence*, 53(22), 26313–26328.
- Breiman, L. (2001). Random forests. *Machine Learning*, 45, 5–32. <http://dx.doi.org/10.1023/A:1010933404324>.
- Carbone, M. R. (2022). When not to use machine learning: A perspective on potential and limitations. *MRS Bulletin*, 47(9), 968–974.
- Dalton, D., Husmeier, D., & Gao, H. (2023). Physics-informed graph neural network emulation of soft-tissue mechanics. *Computer Methods in Applied Mechanics and Engineering*, 417, Article 116351. <http://dx.doi.org/10.1016/j.cma.2023.116351>.
- Fan, H., Jia, Y., Xia, M., Li, H., & Zhang, Z. (2021). 10.3: Numerical simulation of ball-drop on a flexible AMOLED panel. *SID Symposium Digest of Technical Papers*, 52(1), 72–74. <http://dx.doi.org/10.1002/sdtp.14759>.
- Fey, M., & Lenssen, J. E. (2019). Fast graph representation learning with PyTorch geometric. <http://dx.doi.org/10.48550/arXiv.1903.02428>, arXiv preprint arXiv:1903.02428.
- Fortunato, M., Pfaff, T., Wirnsberger, P., Pritzel, A., & Battaglia, P. (2022). Multiscale meshgraphnets. arXiv preprint arXiv:2210.00612.
- Gao, R., Deo, I. K., & Jaiman, R. K. (2024). A finite element-inspired hypergraph neural network: Application to fluid dynamics simulations. *Journal of Computational Physics*, 504, Article 112866. <http://dx.doi.org/10.1016/j.jcp.2024.112866>.
- Grigorev, A., Black, M. J., & Hilliges, O. (2023). Hood: Hierarchical graphs for generalized modelling of clothing dynamics. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 16965–16974).
- He, C., Kong, Q., Ji, K., Xiong, Q., & Yuan, C. (2024). Deep learning-based analysis of interface performance between brittle engineering materials and composites. *Expert Systems with Applications*, Article 124920. <http://dx.doi.org/10.1016/j.eswa.2024.124920>.
- Hwan, C.-L., Lin, M.-J., Lo, C.-C., & Chen, W.-L. (2011). Drop tests and impact simulation for cell phones. *Journal of the Chinese Institute of Engineers*, 34(3), 337–346. <http://dx.doi.org/10.1080/02533839.2011.565608>.
- Incorporated, C. (2020). Corning® gorilla® glass 6 product information sheet. URL: https://www.corning.com/microsites/csm/gorillaglass/PI_Sheets/2020/Corning%20Gorilla%20Glass%206_PI%20Sheet.pdf. (Accessed 13 August 2024).
- Kang, Y.-E., Yang, S., & Yee, K. (2022). Physics-aware reduced-order modeling of transonic flow via β -variational autoencoder. *Physics of Fluids*, 34(7).
- Kim, M. G., Kim, Y., & Kim, Y. M. (2024). Structural sensitivity to reliability of flexible AMOLED modules using mechanical simulation and machine learning. *Organic Electronics*, 125, Article 106967. <http://dx.doi.org/10.1016/j.orgel.2023.106967>.
- Lam, R., Sanchez-Gonzalez, A., Willson, M., Wirnsberger, P., Fortunato, M., Alet, F., et al. (2023). Learning skillful medium-range global weather forecasting. *Science*, 382(6677), 1416–1421. <http://dx.doi.org/10.1126/science.adi2336>.
- Lang, X., Wu, D., & Mao, W. (2024). Physics-informed machine learning models for ship speed prediction. *Expert Systems with Applications*, 238, Article 121877. <http://dx.doi.org/10.1016/j.eswa.2023.121877>.
- Li, S., Robert, A., Faisal, A. A., & Piggott, M. D. (2024). Learning to optimise wind farms with graph transformers. *Applied Energy*, 359, Article 122758. <http://dx.doi.org/10.1016/j.apenergy.2024.122758>.
- Liao, Z., Wang, S., & Komura, T. (2024). SENC: Handling self-collision in neural cloth simulation. <http://dx.doi.org/10.48550/arXiv.2407.12479>, arXiv preprint arXiv:2407.12479.
- Liu, R., Gao, M., Wang, X., Xu, G., Sun, Z., & Zhu, X. (2018). P-10.1: Foldable AMOLED with dropping ball enhancement. vol. 49, In *SID symposium digest of technical papers* (pp. 704–705). <http://dx.doi.org/10.1002/sdtp.12808>.
- Ma, B. S., Jo, W., Kim, W., & Kim, T.-S. (2020). Mechanical modeling of rollable OLED display apparatus considering spring component. *Journal of the Microelectronics and Packaging Society*, 27(2), 19–26. <http://dx.doi.org/10.6117/kmeps.2020.27.2.019>.
- McKay, M. D., Beckman, R. J., & Conover, W. J. (2000). A comparison of three methods for selecting values of input variables in the analysis of output from a computer code. *Technometrics*, 42(1), 55–61. <http://dx.doi.org/10.1080/00401706.2000.10485979>.
- Pan, M.-C., Chen, P.-C., & Ho, J. (2003). Drop simulation and experimental validation of TFT-LCD monitors. In *Proceedings of the 5th electronics packaging technology conference* (pp. 269–274). IEEE, <http://dx.doi.org/10.1109/EPTC.2003.1271534>.
- Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J., Chanan, G., et al. (2019). Pytorch: An imperative style, high-performance deep learning library. *Advances in Neural Information Processing Systems*, 32, <http://dx.doi.org/10.48550/arXiv.1912.01703>.
- Pegolotti, L., Pfaller, M. R., Rubio, N. L., Ding, K., Brufau, R. B., Darve, E., et al. (2024). Learning reduced-order models for cardiovascular simulations with graph neural networks. *Computers in Biology and Medicine*, 168, Article 107676. <http://dx.doi.org/10.1016/j.combiomed.2023.107676>.
- Pfaff, T., Fortunato, M., Sanchez-Gonzalez, A., & Battaglia, P. W. (2020). Learning mesh-based simulation with graph networks. <http://dx.doi.org/10.48550/arXiv.2010.03409>, arXiv preprint arXiv:2010.03409.
- Prokhorenkova, L., Gusev, G., Vorobev, A., Dorogush, A. V., & Gulin, A. (2018). Cat-Boost: unbiased boosting with categorical features. *Advances in Neural Information Processing Systems*, 31.
- Sanchez-Gonzalez, A., Godwin, J., Pfaff, T., Ying, R., Leskovec, J., & Battaglia, P. (2020). Learning to simulate complex physics with graph networks. In *International conference on machine learning* (pp. 8459–8468). PMLR.
- Shao, M., Chen, J., Wang, T., Tao, F., Du, J., Zhang, D., et al. (2023). Accelerating analysis for structure design via deep learning surrogate models. *Advanced Intelligent Systems*, 5(2), Article 2200099. <http://dx.doi.org/10.1002/aisy.202200099>.

- Shin, S., Jin, A.-h., Yoo, S., Lee, S., Kim, C., Heo, S., et al. (2023). Wheel impact test by deep learning: prediction of location and magnitude of maximum stress. *Structural and Multidisciplinary Optimization*, 66(1), 24. <http://dx.doi.org/10.1007/s00158-022-03485-6>.
- Tekin, S. F., Karaahmetoglu, O., Ilhan, F., Balaban, I., & Kozat, S. S. (2021). Spatio-temporal weather forecasting and attention mechanism on convolutional lstms. arXiv preprint [arXiv:2102.00696](https://arxiv.org/abs/2102.00696). 4.
- Toshev, A. P., Paehler, L., Panizza, A., & Adams, N. A. (2023). On the relationships between graph neural networks for the simulation of physical systems and classical numerical methods. arXiv preprint [arXiv:2304.00146](https://arxiv.org/abs/2304.00146).
- Wadagbalkar, P., & Liu, G. R. (2021). Real-time prediction of projectile penetration to laminates by training machine learning models with finite element solver as the trainer. *Defence Technology*, 17(1), 147–160. <http://dx.doi.org/10.1016/j.dt.2020.03.017>.
- Wang, S., Cao, J., & Yu, P. S. (2022). Deep learning for spatio-temporal data mining: A survey. *IEEE Transactions on Knowledge and Data Engineering*, 34(8), 3681–3700. <http://dx.doi.org/10.1109/TKDE.2020.3025580>.
- Wang, K., Tang, X.-Y., & Zhao, S. (2024). Robust multi-step wind speed forecasting based on a graph-based data reconstruction deep learning method. *Expert Systems with Applications*, 238, Article 121886. <http://dx.doi.org/10.1016/j.eswa.2023.121886>.
- Wang, Z., Wang, C., Zhang, S., Qiu, L., Lin, Y., Tan, J., et al. (2024). Towards high-accuracy axial springback: Mesh-based simulation of metal tube bending via geometry/process-integrated graph neural networks. *Expert Systems with Applications*, 255, Article 124577. <http://dx.doi.org/10.1016/j.eswa.2024.124577>.
- Wu, J., Song, G., Yeh, C.-P., & Wyatt, K. (1998). Drop/impact simulation and test validation of telecommunication products. In *ITherm'98. Sixth intersociety conference on thermal and thermomechanical phenomena in electronic systems (cat. no. 98CH36208)* (pp. 330–336). IEEE, <http://dx.doi.org/10.1109/ITHERM.1998.689567>.
- Wu, H., Xu, F., Duan, Y., Niu, Z., Wang, W., Lu, G., et al. (2024). Spatio-temporal fluid dynamics modeling via physical-awareness and parameter diffusion guidance. <http://dx.doi.org/10.48550/arXiv.2403.13850>, arXiv preprint [arXiv:2403.13850](https://arxiv.org/abs/2403.13850).
- Wu, D., Zou, L., Liao, D., Shi, J., Peng, W., & Huang, L. (2022). Numerical simulation theory and structural optimization of a flexible AMOLED module subjected to the impact of a falling ball. *Displays*, 74, Article 102261. <http://dx.doi.org/10.1016/j.displa.2022.102261>.
- Yang, S., Kim, H., Hong, Y., Yee, K., Maulik, R., & Kang, N. (2024). Data-driven physics-informed neural networks: A digital twin perspective. *Computer Methods in Applied Mechanics and Engineering*, 428, Article 117075.
- Yang, S., Lee, S., & Yee, K. (2023). Inverse design optimization framework via a two-step deep learning approach: application to a wind turbine airfoil. *Engineering with Computers*, 39(3), 2239–2255.
- Yang, S., Vinuesa, R., & Kang, N. (2024b). Enhancing graph U-nets for mesh-agnostic spatio-temporal flow prediction. arXiv preprint [arXiv:2406.03789](https://arxiv.org/abs/2406.03789).
- Yang, S., Vinuesa, R., & Kang, N. (2024c). Towards robust spatio-temporal autoregressive prediction: Adams-bashforth time integration with adaptive multi-step rollout. arXiv preprint [arXiv:2412.05657](https://arxiv.org/abs/2412.05657).
- Yang, S., & Yee, K. (2022). Design rule extraction using multi-fidelity surrogate model for unmanned combat aerial vehicles. *Journal of Aircraft*, 59(4), 977–991.
- Zhang, Y.-F., & Thorburn, P. J. (2022). A deep surrogate model with spatio-temporal awareness for water quality sensor measurement. *Expert Systems with Applications*, 200, Article 116914. <http://dx.doi.org/10.1016/j.eswa.2022.116914>.
- Zhao, Y., Li, H., Zhou, H., Attar, H. R., Pfaff, T., & Li, N. (2024). A review of graph neural network applications in mechanics-related domains. *Artificial Intelligence Review*, 57(11), 315.
- Zhao, Y., Liao, H., Pan, S., & Zhao, Y. (2024). Interpretable multi-graph convolution network integrating spatio-temporal attention and dynamic combination for wind power forecasting. *Expert Systems with Applications*, 255, Article 124766. <http://dx.doi.org/10.1016/j.eswa.2024.124766>.